Results Archive

About this archive

This archive accompanies the anonymised submission "What Organizes in Path-Decomposable Neural Network Architectures". It contains 32 figures that we moved out of the supplementary material to keep that material readable. Each figure keeps its original caption and label, and the figures are numbered here from R1 to R32. References without the prefix R, such as Appendix E.7 or Table A.E23, point to the supplementary material of the paper.

The archive follows the three research questions of the paper. Section 1 belongs to RQ.1, which asks what learning organises in the structural disposition and how the optimiser shapes that organisation. Sections 2 to 5 belong to RQ.2, which asks how the disposition is realised when the network is placed under a declared condition. Sections 6 to 10 belong to RQ.3, which asks whether the realised organisation is load-bearing and how the network re-realises a condition after an intervention on it. Each section first states the question that its population answers and the design that answers it. It then reports what the figures establish, with the values and tests on which they rest, and closes with the interpretation and the boundary of its generality. The values are taken from the tables of the supplement and from the analyses of the same experiments, so the archive adds no experiment to the paper.

Reading conventions

States. The structural state uses the measure of the trained network without conditioning on an input. A conditional state uses the measure realised under a condition $c = (x, x^\circ)$. A structural-to-conditional change compares the two measures of one trained network at the same training state, so it is a change of measure and not a change of parameters. On the two benchmarks, the clean state is realised on clean inputs. The shifted state adds Gaussian noise at 15 dB in raw image space and keeps the clean normalisation and the reference $x^\circ = 0$ (Appendix E.9).

Kinds of statement. We distinguish three kinds of statement. An identity holds by construction, such as the exact tilt of Section 4.2 or the sharing decomposition $S_{\text{occ}} = S_{\text{param}} + S_{\text{reuse}}$, and is checked numerically. A recurrence is a direction or an ordering that holds across the units of a population, and it is reported with the number of units that agree. An association is a post-hoc statistical relation, and we give it no causal reading.

Statistics. Population summaries are medians with interquartile bands unless a caption says otherwise. Paired effects are rank-biserial correlations r_{rb} or Hodges-Lehmann shifts. Holm adjustment is applied within each declared family of tests. Kendall's W measures how consistently matched blocks rank the levels of a factor. It does not measure effect size, so we read it together with the contrasts.
Observables. The following symbols recur in the captions. Their full definitions are in Appendices C.1 and A.3.

Symbol	Meaning
η_q	Occupancy of order q: the effective number of paths $N_{\text{eff}}^{(q)}$ divided by the number of architectural paths N_{topo}.
S_{deficit}	Entropy deficit $\log N_{\text{topo}} - H_1$, the distance of the measure from the uniform measure over paths.
ξ_{50}	Characteristic scale at which the box count covers half of the vertices, $N_B/N = 0.5$.
d^*, x_{d^*}	Transition-supported maximum of $d_{\text{eff}}^F = -d \log N_B / d \log x$ on $x = \ell_B/\xi_{50}$, and its location.
Λ_F^*	Peak flow lacunarity: the largest unevenness of flow mass over the boxes.
Inter-box retention	Share of directed flow that survives coarse-graining between boxes.
Quotient-flow KS	Test of whether the coarse flow keeps its shape across resolutions.
Δ_i^F	Local support-mass mismatch $\alpha_i^F - d_i^F$ at node i, fitted on one common window.
D_H, D_{man}	Hellinger displacement of the realised measure, and displacement of the declared response, after intervention.

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1 MLP structural disposition: optimiser

Question. Appendix E.4 establishes that the optimiser sets how far learning concentrates the structural measure. On the 118 matched blocks, the median entropy-deficit increase is 3.323 nats under Adam and 0.062 under SGD (Table A.E17), and the optimiser ordering of this increase has Kendall's $W = 0.902$ (Table A.E18). The exact tilt of Section 4.2 writes each interval of this motion through the covariance of the growth field G with the current log-measure, which factorises into an alignment $\rho_p(G, \log p)$ and an intensity $\mathrm{Var}_p(G)$. On the macro intervals shared by all optimisers, the intensity separates the optimisers far more than the alignment does (Table A.E21). This raises a question about origin. We ask whether the Adam regime is already present in the first update, or whether training generates it.

Design. Figure R1 answers this question with the microstep arm. The arm uses one representative architecture, two maximally separated environments and five seeds. All four optimisers are then measured at the same ten optimiser-step endpoints in each of the ten paired blocks, so every comparison is exactly matched.

The first update does not fix the ordering. At step 1, the median Jeffreys displacement J of the path measure is about 0.0024 under Adam and 0.0216 under RMSProp, so Adam begins with a smaller path-space step than RMSProp. By the interval that ends at step 512, the Adam median has risen to about 0.205 and the RMSProp median to about 0.061. Panels (a) and (c) show that the alignment and the intensity of selection develop over the same steps. Panel (d) adds a second distinction. Early in the arm, the two SGD regimes can move the path measure more per unit of squared parameter displacement than the adaptive methods, although their absolute displacement is small. By step 512, the parameter-normalised values lie much closer together, whereas the absolute displacement under Adam has become far larger.

Interpretation. The Adam regime is therefore generated progressively. It accumulates as a persistent and increasingly coherent path-selective motion, and it does not reduce to larger parameter steps. Parameter displacement and path-measure displacement are therefore distinct coordinates of optimisation, and one cannot be read off the other.

Scope. The arm contains one architecture. It therefore establishes how the regime forms and leaves its architecture dependence to the macro panel, whose dynamic inference is restricted to the six intervals stored identically for every optimiser.

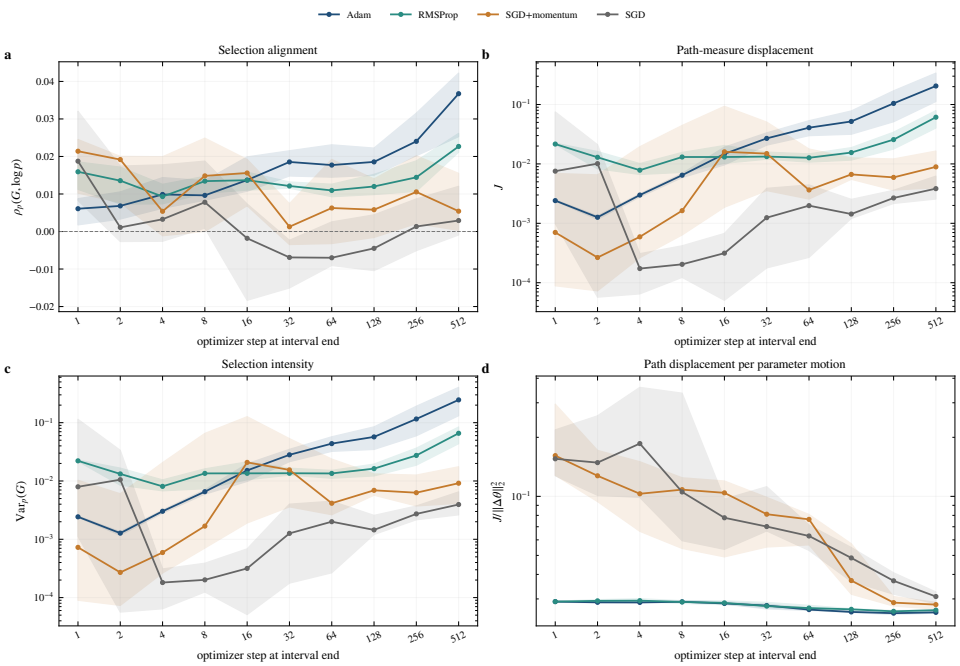


Fig. R1: **Exactly matched microstep dynamics.** Ten paired blocks (one architecture, two environments, five seeds) are measured at the same optimiser-step endpoints. **(a)** Path-selection alignment. **(b)** Jeffreys path-measure displacement. **(c)** Selection intensity. **(d)** Jeffreys displacement normalised by squared Euclidean parameter displacement. Adaptive optimisers do not simply begin with a uniformly larger path-space step; Adam progressively develops much larger absolute path redistribution with exposure. The parameter-normalised panel also shows that this is not equivalent to saying that every unit of Euclidean parameter motion is intrinsically more path-effective under Adam.

2 MLP realisation in the controlled populations

Question. Section 1 concerns the structural measure, into which no input has entered. RQ.2 asks what a declared condition realises from it. The finite-response identity of Section 5.1 gives $Q_t^c(\pi) = Q_t^S(\pi)\, R_{c,t}(\pi)$, so a condition can only reweight the structural support. Whether this reweighting concentrates the measure, and which structural orderings survive it, are empirical questions. Appendix E.7 answers them on four synthetic MLP populations, with designs and tests in Tables A.E23 to A.E28. Inputs are nested within trained runs, so the inferential unit is the matched model block throughout.

Architecture survives as an ordering of probability mass. Figure R2 follows the architecture cohort of 20 architectures with eight seeds, trained with Adam under one environment, together with a six-architecture tanh sentinel. Over training, the conditional entropy deficit grows by a median 8.784 nats against 2.441 for the structural

deficit, with the same sign in all 20 architectures, and the KL departure of the conditional from the structural measure grows by 8.436 (panel a). At the best-validation state, architecture orders the structural deficit with $W_A = 0.982$ and the conditional deficit with $W_A = 0.978$, while the downstream geometry forms a lower tier, from 0.960 for peak lacunarity to 0.708 for ξ_{50} (Table A.E24a; panel b). The seedwise structural-to-conditional architecture bridge separates these two kinds of observable (panel c). It is 0.965 for the entropy deficit and 0.689 and 0.605 for edge and node heterogeneity. For ξ_{50}, d^*, peak lacunarity and retained flow it is -0.041, -0.398, 0.498 and -0.447. Panel (d) shows when these orderings form. The conditional deficit and the KL departure keep much of their final architecture hierarchy from initialisation, with correlations of 0.925 and 0.951, whereas flow heterogeneity and most geometric rankings form during training.

Architecture therefore fixes the carrier and organises the scale of concentration, while the finite condition decides how that scaffold is used. The geometry is where this use becomes visible. Conditioning produces architecture-specific plateaus and coarse transitions in the covering response (panel e), and a credible scaling window, present in all 160 structural states, survives in 48 conditional ones (Table A.E25). Panel (f) marks two boundaries. ReLU realises much stronger concentration than the tanh sentinel, by 10.030 nats in conditional deficit. Moreover, the geometry of the population-average flow differs from the geometry that a typical single input realises.

The environment keeps its effect and loses its ordering. Section 4.1 shows that the environment modulates the magnitude of structural concentration while the architecture ordering stays almost fixed. Figure R3 asks whether the environment keeps its own ordering once computation is realised. The matrix crosses 24 frozen environments with six architectures, two seeds and two paired tasks. The deformation recurs without exception. In both tasks, all 24 environments agree in direction for the deficit, both occupancies and both flow heterogeneities, and the median deficit rises by 3.140 nats under regression and 2.766 under classification (Table A.E27a; panel b). The environment remains strongly identifiable ($W_E = 0.548$ and 0.687; panel a), yet none of the core structural-to-conditional environment-rank correlations survives the Holm family.

The task acts as a second transformation of the realised measure. In matched environments, classification is less concentrated than regression, by -0.428 in conditional deficit and -0.746 in KL departure (panel c). Across the two tasks, the environment ordering of concentration is largely kept ($\rho_s = 0.693$ for the conditional deficit and 0.730 for η_1), whereas the ordering of the KL departure reverses ($\rho_s = -0.431$; panel d). Realisation also separates the tasks more strongly than the weights do. Under ReLU, the median Hellinger distance between matched regression-trained and classification-trained measures rises from 0.375 structurally to 0.756 under conditioning, with an increase in all 24 environments (panel e). Signal-to-noise ratio is the clearest modulator of this task contrast, with a correlation of about -0.872 with the KL task contrast (panel f).

Figure R4 excludes the simplest clock explanation. It compares environments at matched fractions τ_L of validation-loss improvement. Under regression, the environment effect on the conditional deficit falls from $W_E = 0.759$ at $\tau_L = 0.25$ to 0.473 at

$\tau_L = 0.99$ and remains substantial. Under classification, the environment effect on the KL departure moves from 0.669 to 0.526 over the same range (panels a and b). Internal organisation also continues after most of the functional progress. From the first scheduled state at or beyond 90% of the improvement to epoch 280, the regression medians still rise by 0.626 in conditional deficit and by 1.625 in KL departure (panel c). Panel (d) relates post-best functional deterioration to signal-to-noise ratio, and panel (e) shows that conditioning removes the finite-scaling regime in a non-negligible subset of states. The environment therefore acts on realisation as a field. It modulates which structural possibilities are learned and how strongly a finite condition selects among them, and the direction of its action recurs while its ordering is free to change.

The optimiser phenotype survives, and its ordering is relative. Section 4.2 shows that the optimiser sets the magnitude of structural concentration. Figure R5 asks whether this phenotype, and its ordering, survive realisation. The panel crosses six architectures, two environments, three seeds, two activation classes and four optimisers. The optimiser becomes more visible under realisation. Its concordance rises from 0.282 for the structural deficit to 0.706 for the conditional deficit and 0.787 for the KL departure (Table A.E27b; panel a). Its ordering changes. RMSProp realises the largest conditional deficit, 7.414 against 5.575 for Adam, although Adam is the strongest structural concentrator (panel b). The cumulative realised Hellinger motion separates the regimes most clearly ($W_O \approx 0.912$), with medians of about 4.63 under RMSProp and 2.80 under Adam (panel c).

Activation decides how much of the structural ordering is inherited (panel d). Under ReLU, the optimiser-rank bridge stays positive, at 0.50 for the deficit and 0.80 for node heterogeneity. Under tanh, many bridges weaken or reverse, to about -0.20 for the deficit and -0.50 for edge heterogeneity. Matching functional progress leaves the effect in place, with $W_O \approx 0.708$ for the conditional deficit at $\tau_L = 0.25$ and 0.660 near $\tau_L = 0.99$ (panel e). The downstream geometry attenuates it. At epoch 280, the optimiser effect is about 0.506 for peak lacunarity and retained flow, 0.296 for ξ_{50} and 0.023 for quotient-flow KS (panel f). The optimiser therefore changes a trained state whose expression under a condition depends on the activation response, and its rank is relative to the measure being observed.

Transition. In MLPs, every coefficient is executed once. Section 3 asks how these relations change when one coefficient is executed at many positions.

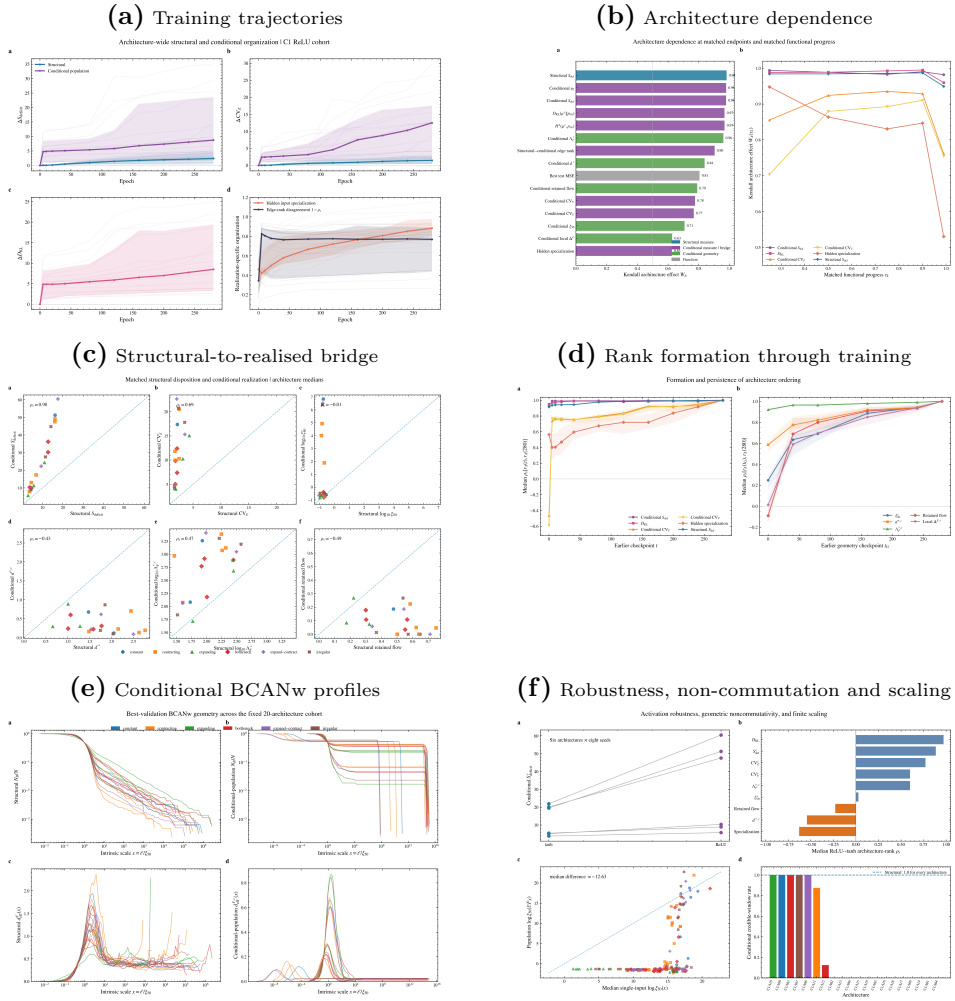


Fig. R2: Architecture cohort under conditional realisation. **(a)** Structural and conditional entropy deficit share the training direction but differ strongly in amplitude, while realised edge-flow heterogeneity, KL departure and input specialisation continue to develop. **(b)** Architecture dependence remains strongest for concentration and structural-to-realised displacement after matching functional progress. **(c)** Conditional concentration is strongly rank preserved and amplified relative to the structural disposition, while characteristic scale, crossover, lacunarity and retained coarse flow are progressively reorganised. **(d)** Conditional concentration and KL retain an early architecture scaffold, whereas edge and node heterogeneity and most conditional geometric rankings form during training. **(e)** Conditioning generates architecture-specific BCANw plateaus and coarse transitions rather than a uniform displacement of the structural covering family. **(f)** ReLU realises stronger concentration than the tanh sentinel, population and typical-input geometry differ, and finite conditional scaling windows are architecture dependent.



Fig. R3: Environment matrix: environment and task organisation of conditional realisation. **(a)** Environment remains strongly expressed in conditional concentration, structural departure, flow heterogeneity, specialisation and several BCANw coordinates. **(b)** The structural-to-realised deformation recurs in sign across environments more strongly than environment rank is preserved. **(c)** Regression and classification task changes are coherent across matched environments. **(d)** Several realised coordinates preserve cross-task environment rank, whereas KL departure shows a significant reversal. **(e)** Conditional realisation amplifies the separation of the regression-trained and classification-trained states beyond their structural distance. **(f)** Signal-to-noise ratio is the clearest environmental coordinate modulating the task contrast.

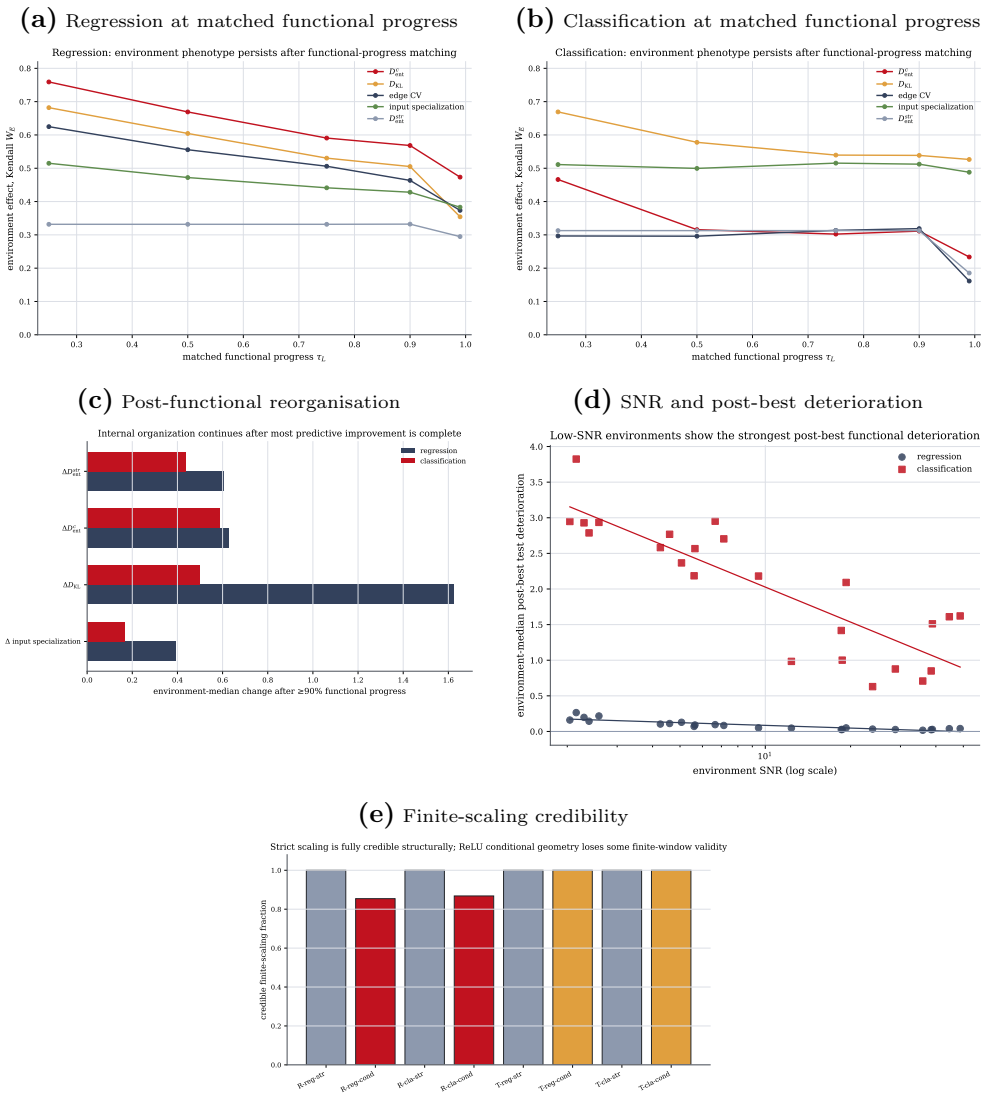


Fig. R4: Environment matrix: temporal separation and reliability. **(a,b)** Environment effects remain substantial after matching the functional-progress clock under regression and classification. **(c)** Structural and conditional concentration, KL departure and specialisation continue to reorganise after most predictive improvement has occurred. **(d)** Low-SNR environments show stronger post-best functional deterioration in both paired tasks. **(e)** Structural finite-scaling windows remain broadly admissible, whereas conditional realisation removes the valid finite-scaling regime in a non-negligible subset of states.

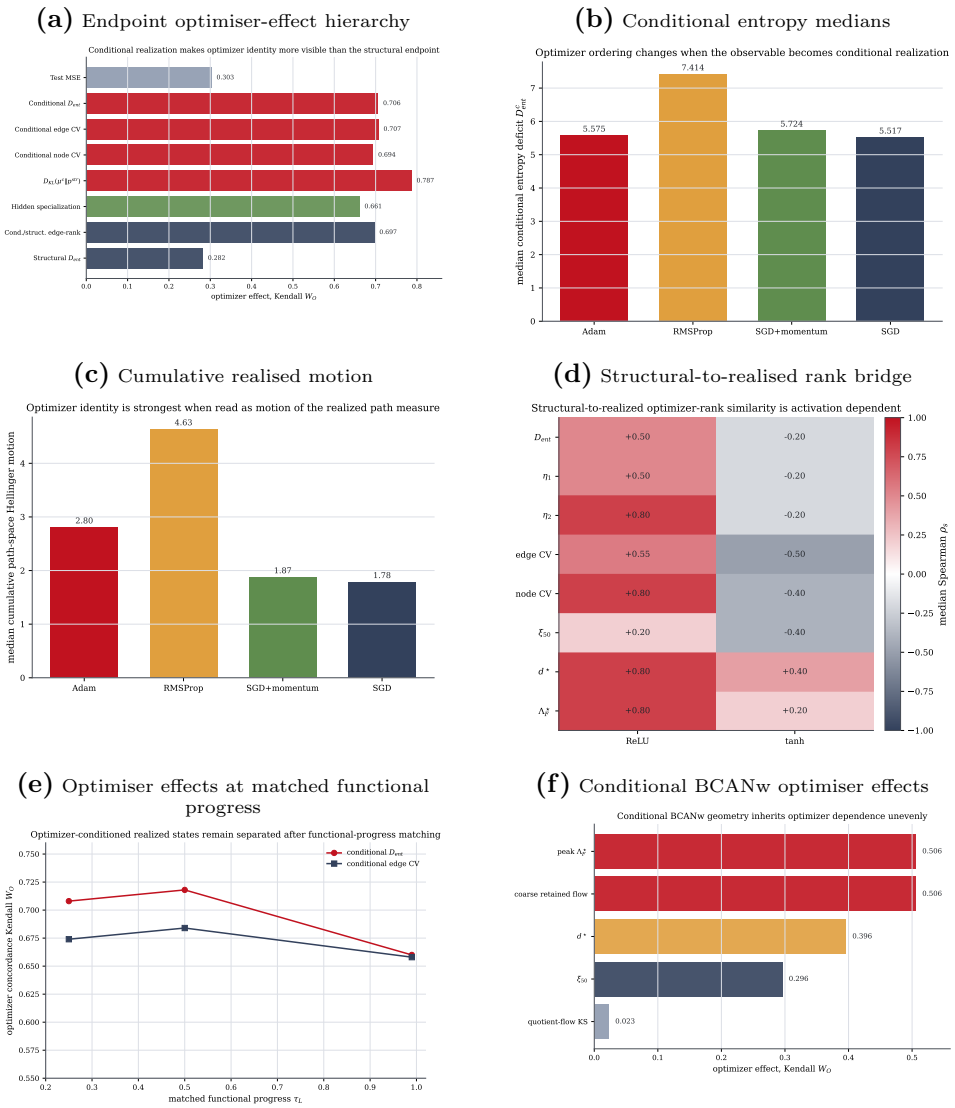


Fig. R5: Optimiser panel under conditional realisation. **(a)** Optimiser identity is more visible in conditional concentration, structural departure, flow heterogeneity and specialisation than in the structural entropy endpoint. **(b)** The conditional entropy ordering differs from the structural ordering. **(c)** Cumulative realised Hellinger motion separates the optimiser regimes over training. **(d)** ReLU retains a positive structural-to-realised optimiser-rank bridge for several descriptors, whereas tanh weakens or reverses many rankings. **(e)** Conditional concentration and edge-flow heterogeneity remain optimiser dependent after matching predictive progress. **(f)** Downstream BCANw geometry attenuates the proximal optimiser effect, with lacunarity and retained coarse flow more sensitive than characteristic scale or quotient-flow KS.

3 CNN realisation in the controlled populations

Question. A convolutional coefficient is executed at every compatible position, so the occurrence deficit of a CNN decomposes exactly as $S_{\mathrm{occ}} = S_{\mathrm{param}} + S_{\mathrm{reuse}}$, for the conditional measure as for the structural one. Structurally, learning raises S_{occ} almost entirely through S_{param}, by 0.682 of a median rise of 0.685 (Section 4.2). Appendix E.8 asks which term carries the conditional selection, and whether the architecture and optimiser orderings survive it.

Global concentration keeps the architecture order, and sharing reorganises it. Figure R6 follows the environment panel, which crosses the 24 frozen environments with 12 architectures and two seeds. The environment is the inferential unit for environment claims. The deformation recurs in every environment. The median path entropy falls by 2.903 nats in all 24 environments, S_{occ} and S_{reuse} rise in all 24, and S_{param} falls in 23 (Table A.E31a; panel c). Panel (a) separates two kinds of architecture memory. The global concentration observables keep the structural architecture ordering under conditioning, with median rank correlations from 0.909 to 0.958 and values above 0.7 in every environment. The sharing components do not keep it, with medians of 0.469 for S_{occ}, 0.031 for S_{param} and 0.406 for S_{reuse}. Panel (b) places this beside the structural result of Appendix E.5. The global order is stable across environments structurally and also survives conditioning within each environment, whereas the sharing-level order loses this commutation under realisation.

The geometry deforms with the same regularity (panel d). Within the 24 environments, d^* and crossover lacunarity fall in all 24, inter-box retention rises in all 24, and only strict d_B^F has no recurrent direction (Table A.E31b). The environment modulates the magnitude of realisation more clearly than it modulated the structural law. Interaction strength correlates with the conditional deficit at $\rho_s = 0.782$ after Holm adjustment (Table A.E31c), and we read this as a descriptive relation over the designed panel.

The optimiser alters the transformation itself. Figure R7 follows the compact optimiser panel of 30 matched blocks. Structurally, the optimiser hardly orders the global quantities ($W = 0.037$ for the deficit), and only S_{reuse} is ordered after correction ($W = 0.240$). Under conditioning, the concordance rises to 0.585 for the deficit and 0.631 for S_{reuse}. The structural-to-conditional deformation is itself optimiser dependent, with $W = 0.568$ and 0.648 (Table A.E29d; panel a). The structural order is only weakly inherited (panel b). Within blocks, the median rank correlation between the four structural and the four conditional optimiser values is about 0.20 for the deficit and -0.40 for S_{reuse}, and the exact order is matched in only about 3 to 23% of blocks. The conditional path entropy separates the optimisers from the first epochs and then follows optimiser-specific trajectories (panel c).

Interpretation. In CNNs, realisation preserves the global architecture phenotype and reorganises its implementation through sharing. In the architecture panel, the median rank retention of parameter identities is 0.803, against 0.024 for occurrences (Section 5.2). The learnt coefficients thus remain recognisable, while the spatial sites at which they carry flow are selected anew by the condition. The optimiser acts on this selection most strongly through the reuse channel. The optimiser panel is smaller than the structural one, so its p-values are not compared with those of the structural panel.

Transition. The controlled populations are trained on synthetic teacher tasks. Sections 4 and 5 test the same relations on real images.

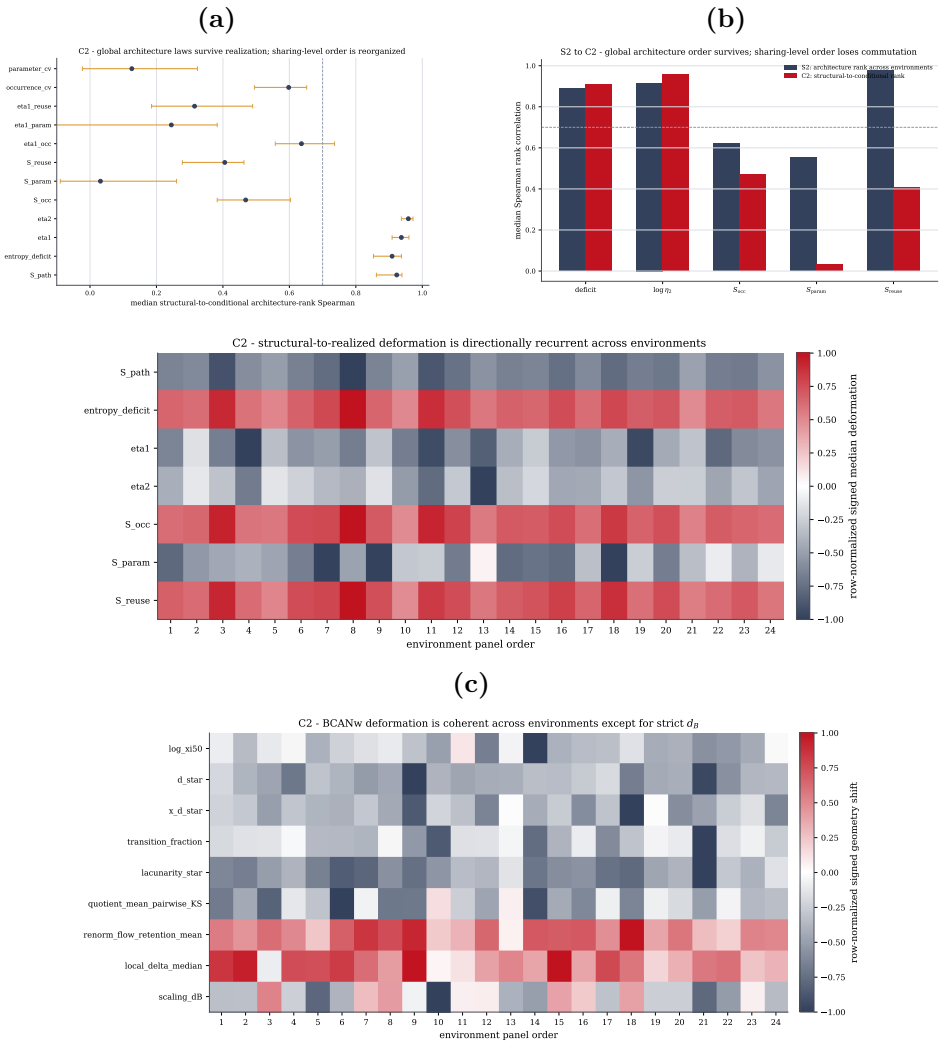
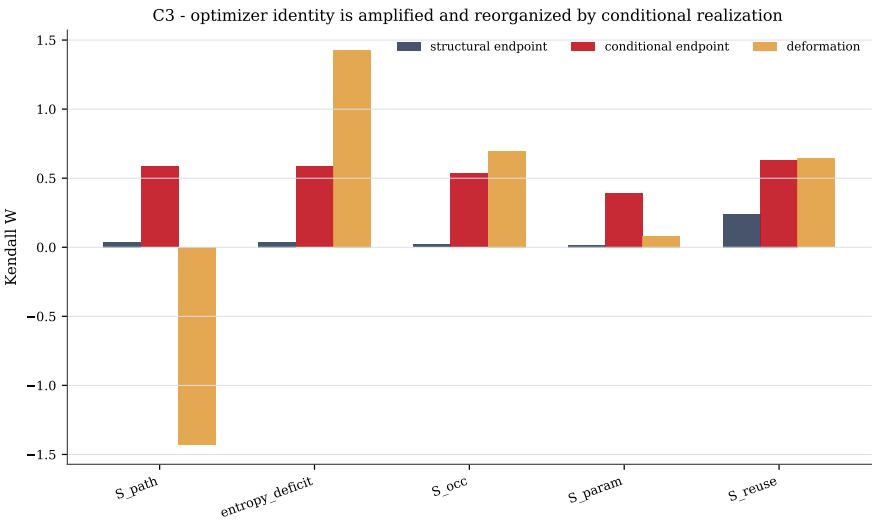
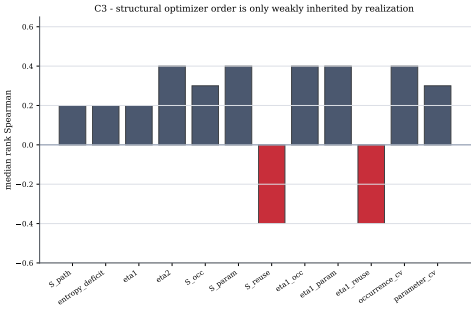


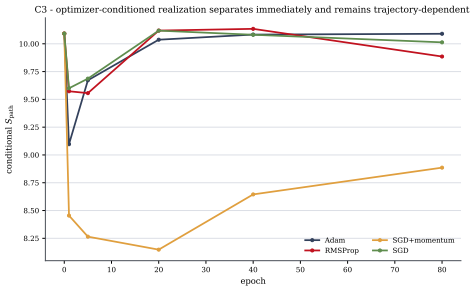
Fig. R6: CNN environment panel under conditional realisation. **(a)** Global complete-path quantities preserve the architecture ordering strongly across the 24 environments, while occurrence, parameter and reuse specialisations are reorganised. **(b)** Two rank-invariance questions: whether the architecture ordering is stable across environments structurally, and whether the structural architecture ordering survives conditioning within each environment. Global concentration is robust in both senses, while sharing-level ordering loses commutation under realisation. **(c)** Environment-by-deformation field for the intrinsic observables; rows are normalised for visualisation only, and the tests use the unnormalised environment medians. **(d)** Environment-by-BCANw deformation field; the direction of geometric deformation recurs across environments for all headline observables except strict d_B^F.



(a)



(b)



(c)

Fig. R7: CNN optimiser panel under conditional realisation. **(a)** Matched optimiser effect sizes at the structural endpoint, at the conditional endpoint and for the structural-to-conditional deformation; conditioning amplifies optimiser dependence, especially in the reuse channel. **(b)** Low and sometimes negative rank correlations between the structural and conditional optimiser orders show that conditioning reorganises the optimiser phenotype. **(c)** The conditional path entropy develops along optimiser-specific trajectories.

4 Case study: MPI3D

Question and design. The benchmark case studies ask whether the relations of the controlled populations hold when the input distribution is not designed. One factorial is trained on each benchmark. Three MLPs and three CNNs are crossed with SGD with momentum, RMSProp and Adam, with ReLU, tanh and SiLU and with three seeds, giving 162 networks. MPI3D is posed as regression of the horizontal position of an object over 40 levels, and its other generative factors act as structured nuisance. Each network is observed with fixed parameters in the structural, clean and shifted states. The weights, endpoint population, ruler, scale grid and detectors are held fixed, so the clean-to-shifted contrast changes only the sensed condition. Intrinsic observables use 160 shared endpoint conditions, and geometry uses a preregistered cohort of 30 networks per state. Every geometry passes the construction gates, and the strict finite-window criterion holds for 96.7%, 90.0% and 86.7% of the structural, clean and shifted geometries. The paper keeps the summary figure (Figure A.E27), and the tests are in Tables A.E34 to A.E39.

Clean evidence sharpens realisation, and degraded evidence diffuses it. Figure R8 shows that clean conditioning contracts the effective support by about two orders of magnitude. The median η_1 falls from 5.92×10^{-3} to 4.02×10^{-5}, which is 0.68% of its structural value, and every network moves in this direction ($r_{rb} = -1.000$; Table A.E35). The contraction holds within ReLU, tanh and SiLU separately. The sensor shift then spreads realised mass over more of the structural support. Relative to the clean state, η_1 rises ($r_{rb} = +0.910$) and the normalised deficit falls ($r_{rb} = -0.927$). The same trained network thus admits a wide disposition and realises a small fraction of it for a specific input, and a controlled degradation of that input widens the realised fraction again.

Conditioning reorganises the factor hierarchy. Figure R9 shows that the activation becomes the most consistently ordered factor after conditioning. Its concordance for η_1 rises from 0.498 to 0.849 under clean and 0.838 under shifted conditions, whereas the optimiser concordance of the realised states stays between about 0.18 and 0.24 (Table A.E36). The pairwise contrasts locate this rise. Structurally, the ReLU-SiLU contrast of the normalised deficit is not separable (Hodges-Lehmann -0.00227, Holm $p = 0.596$). Under clean conditioning, it becomes positive and highly concordant (Hodges-Lehmann $+0.0505$, $r_{rb} = +0.809$, Holm $p = 2.34 \times 10^{-7}$). This is the reordering expected when the finite-response factor R_c, through which the activation law enters, multiplies the structural score. The architecture responds by carrier. For MLPs, its concordance for the normalised deficit falls from 0.635 to 0.203, while for CNNs its concordance for η_1 rises from 0.553 to 0.759.

Realised flow is patchier and more weakly coupled across coarse boxes. Figure R10 shows that the structural-to-clean change flattens the scale response and collapses its transition support. The median d^* falls from 1.635 to 0.501 and the transition fraction from 0.844 to 0.563, while ξ_{50} moves from 0.313 to 0.403 (Table A.E34). Figure R11 shows what replaces the lost scale support. Peak lacunarity rises from 445.2 to 1105.8 and inter-box retention falls from 0.325 to 0.065, with $r_{rb} = +1.000$ and -1.000 (Table A.E35). The Rényi quotient-flow occupancies are normalised by the number of available relations at each scale, so their redistribution does not follow

from the box count alone. These estimators describe the concentration and the inter-box coupling of the represented flow, and we do not read them as a decomposition of the network into literal channels.

The shift deforms the geometry with a definite sign. Figure R12 pairs each geometry network with itself. Under the shift, ξ_{50} contracts ($r_{rb} = -0.755$), d^* rises ($r_{rb} = +0.669$) and peak lacunarity falls ($r_{rb} = -0.987$), all after Holm adjustment. The intrinsic and the geometric arms are defined independently, and they agree here in direction: the shifted realisation occupies more paths, and its representation becomes less patchy and richer in transitions. The evidence supports a correspondence of population-level signs. It does not establish a one-to-one map from the intrinsic change of a network to its geometric change.

Local mismatch is a relation between the task and the carrier. Figure R13 resolves the local mismatch by family. The clean per-geometry medians are -0.248 for MLPs and $+0.341$ for CNNs, with structural-to-clean displacements of -0.258 ($r_{rb} = -0.917$) and $+0.385$ ($r_{rb} = +1.000$; Table A.E37a). Pooled, the clean median is $+0.004$ (Table A.E34), so pooling almost cancels a large opposition. In clean MLPs, realised mass accumulates more slowly than metric support expands over the accepted window, and in clean CNNs it accumulates faster. Under the shift, both families move back towards zero from opposite sides. Panel (d) records a methodological fact. Averaging the flow before the geometry is built and averaging the geometry after conditioning are different estimands, $\mathcal{G}_R(\mathbb{E}_c[F^c]) \neq \mathbb{E}_c[\mathcal{G}_R(F^c)]$. The inverse-flow ruler sends near-zero single-input flows to very long edges, so single-input values of ξ_{50} can lie many orders of magnitude from the population-first value, and we do not read them as length scales.

Realisation selects where shared parameters are used. Figure R14 repeats the controlled CNN result on real data. Across the CNN layer states, the median occurrence deficit rises from 1.245 to 3.134 under clean conditioning, the parameter-identity deficit moves only from 0.937 to 0.913, and the reuse deficit rises from 0.139 to 2.210 (Table A.E37b). The effective occurrence fraction contracts from 0.288 to 0.045, while the parameter fraction stays near 0.40. The reuse deficit is a residual under the declared layer decomposition, and we do not read it as a mutual information.

On MPI3D, realised occupancy carries performance information. Figure R15 asks whether occupancy adds to the design factors in accounting for log test error. The factors alone give $R^2 = 0.8145$. Structural occupancy adds 0.0056 ($p = 0.054$), whereas clean occupancy adds 0.1063, with coefficient 0.151 and 95% interval $[0.125, 0.177]$ (Table A.E38). Since both variables are logarithmic, a tenfold increase of clean η_1 corresponds under this model to about 1.42 times the test error, with architecture, optimiser and activation held fixed. The model is post hoc, associative and cluster-robust over 54 design cells, and it is not a mediation analysis. Section 5 tests whether this association holds on a second task.

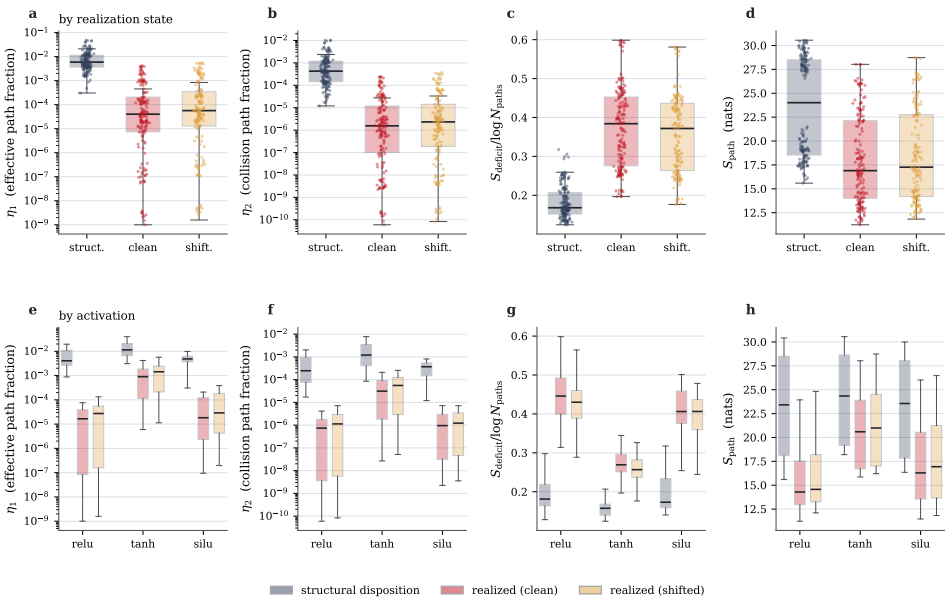


Fig. R8: MPI3D: intrinsic realisation landscape over the 162 networks. Top row: distributions of η_1, η_2, normalised entropy deficit and path entropy by state. Bottom row: the same observables resolved by activation. Each point is one network, after the median over the shared 160-endpoint population; η axes are logarithmic. Clean conditioning contracts effective support by about two orders of magnitude while increasing normalised concentration, and the ordering holds within ReLU, tanh and SiLU separately. The figure concerns occupancy of the conditional measure and makes no statement about predictive quality.

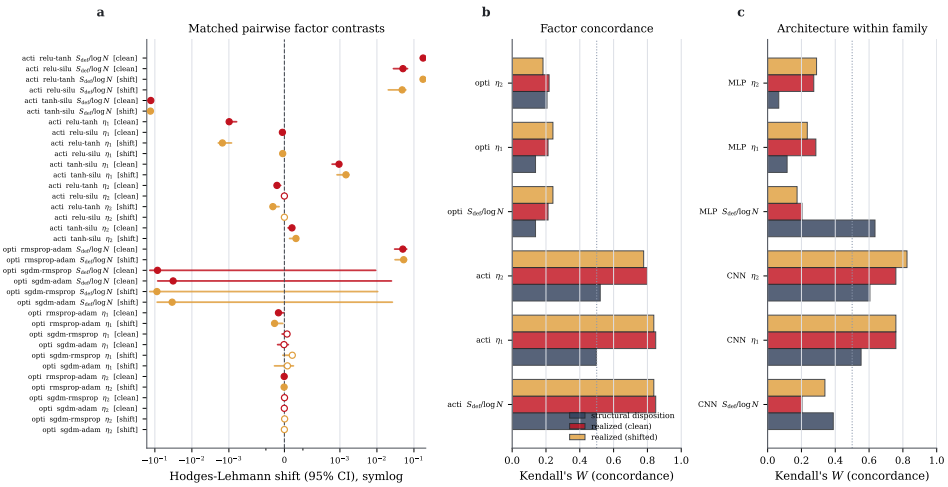


Fig. R9: MPI3D: reorganisation of the factor hierarchy under conditioning. **(a)** Hodges-Lehmann pairwise factor contrasts with distribution-free 95% intervals; filled points are Holm significant. **(b)** Kendall's W for activation and optimiser. **(c)** Architecture concordance within the MLP and CNN families. Matched blocks fix all other design factors and the seed. Conditioning raises activation concordance substantially, while architecture responds differently in the two carriers. W measures ranking concordance and not effect magnitude, so it is read together with the contrasts.

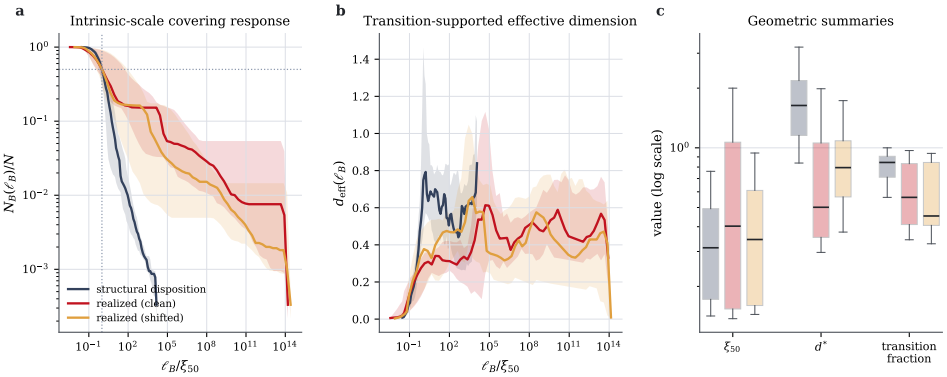


Fig. R10: MPI3D: intrinsic-scale covering response. **(a)** $N_B(\ell_B)/N$ against ℓ_B/ξ_{50}, with the half-cover reference. **(b)** Transition-supported d_{eff}^F. **(c)** Per-geometry summaries. Lines are medians with interquartile bands. The main structural-to-clean change is a flattening of the scale response and a collapse of transition support, not a rescaling of ξ_{50}. The curves are ruler-dependent representations of the flow, and no global dimension is asserted for states that do not support one.

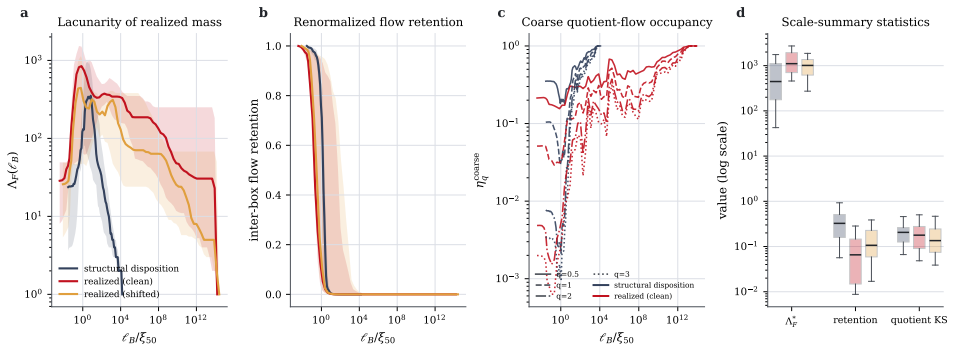


Fig. R11: MPI3D: multiscale texture and coarse flow. **(a)** Flow-mass lacunarity. **(b)** Inter-box flow retention. **(c)** Normalised quotient-flow Rényi occupancies for $q \in \{0.5, 1, 2, 3\}$. **(d)** Per-geometry peak and scale summaries. Clean realised flow is markedly more lacunar and retains much less flow between coarse boxes than the structural disposition. The Rényi occupancies are normalised by the number of available quotient relations at each scale, so their change does not follow from the box count alone.

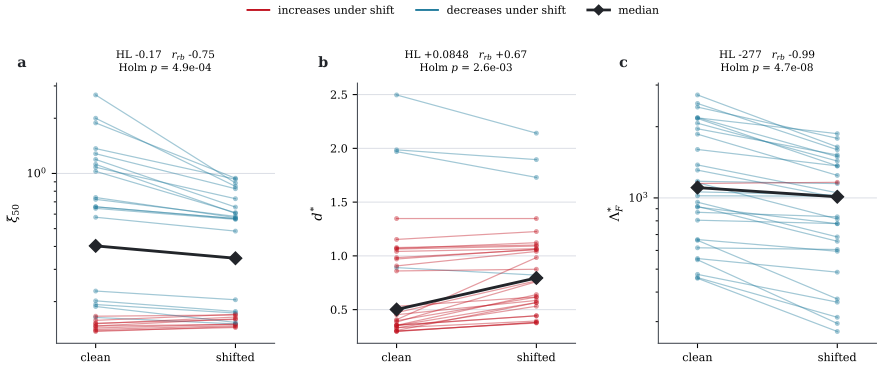


Fig. R12: MPI3D: paired geometric deformation under the 15 dB sensor shift. One line per network joins its clean and shifted values; the heavy line is the median and colour gives the sign of the change. Titles report the Hodges-Lehmann shift, rank-biserial effect and Holm-adjusted Wilcoxon p. **(a)** ξ_{50}. **(b)** d^*. **(c)** Λ_F^*. The deformation is strongly directional and agrees qualitatively with the intrinsic de-concentration under the same perturbation.

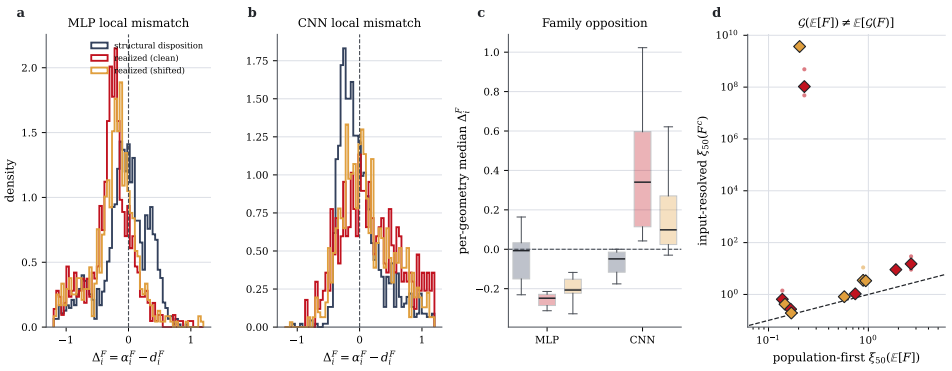


Fig. R13: MPI3D: carrier-resolved local support-mass mismatch and geometric non-commutation. **(a,b)** Node-level $\Delta_i^F = \alpha_i^F - d_i^F$ for the MLP and CNN carriers. **(c)** Per-geometry local medians, revealing the clean MLP and CNN sign opposition that pooling almost cancels. **(d)** Input-resolved against population-first ξ_{50}; diamonds mark within-cell input medians. Both local exponents use the same accepted radial window per node. The large deviations in (d) reflect the nonlinearity of the geometric functional under sparse single-input flow and are not read as a universal length scale.

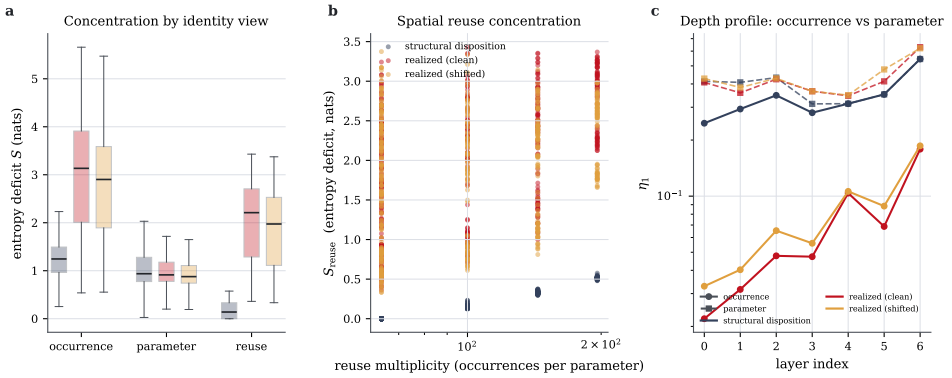


Fig. R14: MPI3D: occurrence, parameter-identity and spatial-reuse concentration in CNNs. **(a)** Entropy deficits in the three views. **(b)** Reuse deficit against the exact spatial reuse multiplicity. **(c)** Depth profiles of η_1 for the occurrence and parameter views. Both distributions derive from the same realised flow and are related by the sharing pushforward. Clean conditioning strongly concentrates spatial occurrence while leaving parameter-identity occupancy comparatively stable. The reuse deficit is a residual under the declared layer decomposition and is not read as a mutual information.

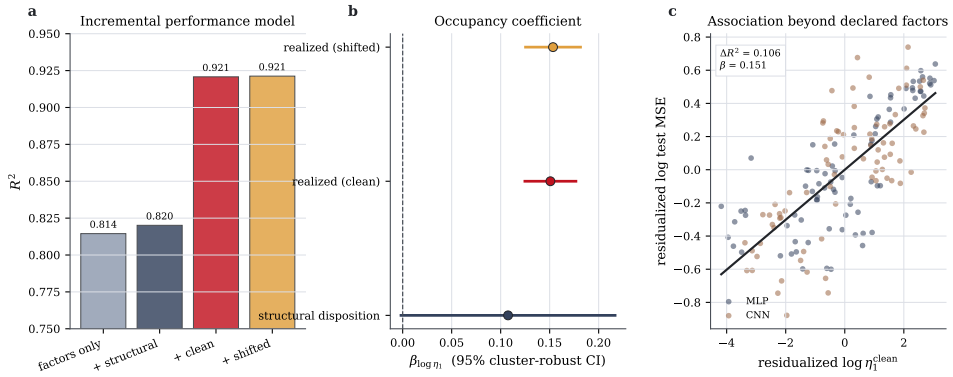


Fig. R15: MPI3D: post-hoc association between occupancy and regression error. **(a)** R^2 of the design-factor model and of the three state-specific augmented models. **(b)** Coefficient of $\log \eta_1$ with 95% cluster-robust intervals. **(c)** Partial-regression view of clean occupancy after architecture, optimiser and activation are residualised from both axes. Clean and shifted occupancy add about 0.106 to R^2 and structural occupancy 0.006. The model is post hoc, associative and cluster-robust over 54 design cells, and it is not a mediation analysis.

5 Case study: Fashion-MNIST

Question and design. Fashion-MNIST changes the task to ten-class classification on a natural image distribution. The architectures, optimisers, activations, seeds, endpoint population, shift, geometry cohort and BCANw implementation are those of Section 4, so the two benchmarks can be aligned directly. Over the 162 networks, the median best test cross-entropy is 0.451 and the median accuracy 0.849. Every geometry satisfies the credibility criterion. The figures follow the MPI3D order, so Figure R16 corresponds to Figure R8, and so on. We read them as a test of which MPI3D relations recur and which depend on the task, and Table A.E39 records the outcome.

The selection of the disposition recurs. Clean conditioning again contracts the support in every network, to a median η_1 of 7.82% of its structural value (Figure R16). ReLU realises the most concentrated measure, tanh the least and SiLU lies between them. The factor hierarchy is reorganised in the same way as on MPI3D (Figure R17). The activation concordance for η_1 rises from 0.700 to 0.931, the optimiser concordance falls from 0.285 to 0.108, and the architecture concordance for η_1 falls from 0.531 to 0.236 in MLPs and rises from 0.790 to 0.931 in CNNs (Table A.E36). The covering response again loses its transition support (Figure R18). The median d^* falls from 2.919 to 1.045 and the transition fraction from 0.922 to 0.414, while ξ_{50} moves only from 0.306 to 0.290. The realised flow is again more lacunar and less inter-coupled (Figure R19), since peak lacunarity rises from 178.0 to 319.2 and retention falls from 0.682 to 0.358. The quotient-flow KS also rises from 0.155 to 0.192 ($r_{rb} = +0.729$), so the coarse flow changes its shape more across resolutions. In CNNs, the reuse deficit rises from 0.131 to 1.049 and the effective reuse fraction falls from 0.877 to 0.357, while the parameter-identity deficit changes by less than 0.15 (Figure R22; Table A.E37b). The reuse mechanism therefore recurs on both tasks.

The response to the shift depends on the task. On Fashion-MNIST, the shift concentrates the realised measure further, with $r_{rb} = -0.986$ for η_1, which is the opposite sign to MPI3D. Its geometric trace is also selective (Figure R20). Only ξ_{50} changes after correction, by $+1.52 \times 10^{-3}$ ($r_{rb} = +0.665$), while d^* ($r_{rb} = +0.234$) and peak lacunarity ($r_{rb} = -0.187$) do not change detectably. A condition can therefore redistribute path mass very consistently without forcing every geometric projection of that mass to change with comparable magnitude. The intrinsic and the geometric arms are related, and they are not interchangeable measurements.

The local sign depends on the task and the carrier. In Figure R21, the structural MLP median of the local mismatch is +0.100 and moves to +0.043 under clean realisation ($r_{rb} = -0.900$, Holm $p = 1.7 \times 10^{-3}$). The CNN median is −0.108 and moves to −0.079 ($r_{rb} = +0.633$, Holm $p = 0.060$). Both families thus approach zero from opposite sides, and each carrier has the opposite clean sign to its MPI3D counterpart. The local mismatch is therefore an informative relational diagnostic, and its sign is not a universal invariant of realisation. Across the input-resolved sentinel, the median ratio between the single-input and the population-first ξ_{50} is about 3.28 under clean and 3.40 under shifted conditions. One ReLU MLP case reaches a ratio of about 3.5×10^9, which we read as sparse-flow stretching under the inverse-flow ruler.

Realised occupancy is not a universal performance surrogate. In Figure R23, the design factors alone explain $R^2 = 0.8515$ of log test cross-entropy. Structural, clean

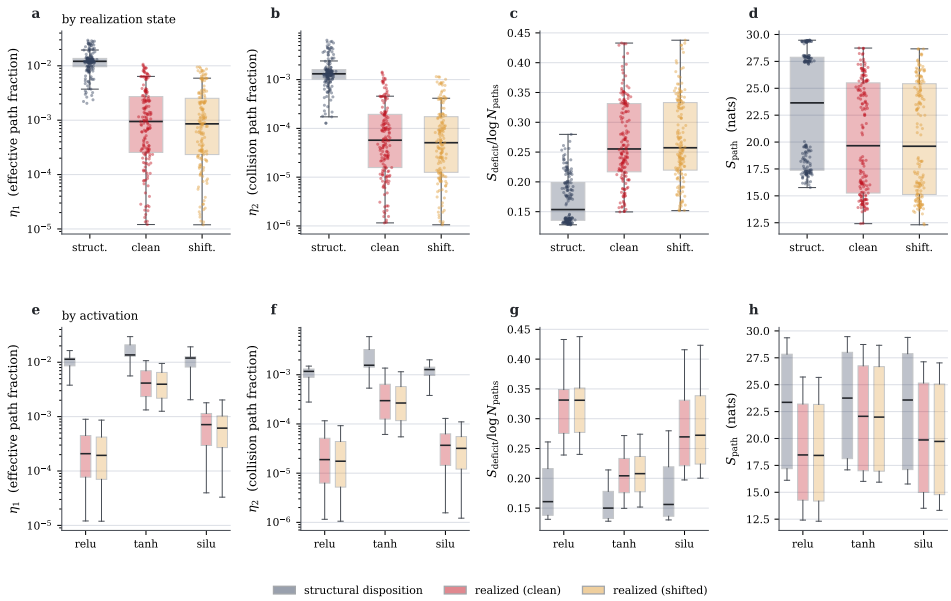


Fig. R16: Fashion-MNIST: intrinsic realisation landscape. **(a)** to **(d)** Distributions of η_1, η_2, normalised entropy deficit and path entropy in the structural, clean and shifted states. **(e)** to **(h)** The same observables by activation. The structural-to-clean contraction holds for all activations; ReLU realises the most concentrated measure, tanh the least, and SiLU lies between them.

and shifted occupancy add 0.0016, 0.0049 and 0.0037, and no occupancy coefficient excludes zero (clean coefficient -0.0146, $p = 0.102$; Table A.E38). A model of log classification error reaches the same conclusion. The strong MPI3D association is therefore a property of that task, and not a defining property of conditional realisation.

Synthesis. The two benchmarks agree on the existence and the selectivity of realisation. On both, the condition contracts the support, strengthens the activation ordering, collapses the transition support of the geometry and concentrates CNN computation in spatial reuse. They differ in the sign of the response to the shift, in the sign of the local mismatch and in the informativeness of occupancy for performance. Section 6 turns from description to intervention and asks whether the realised part of the computation is load-bearing.

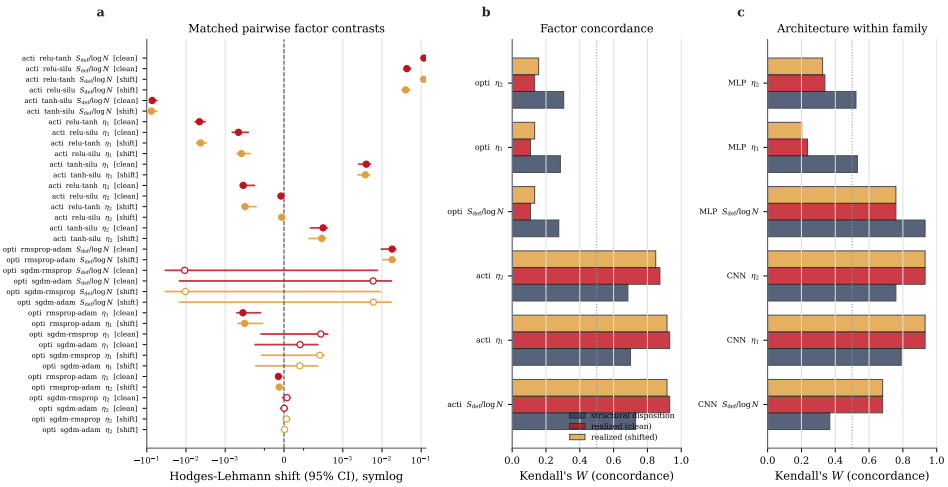


Fig. R17: Fashion-MNIST: factor hierarchy under structural and conditional observation. **(a)** Matched pairwise Hodges-Lehmann contrasts with bootstrap 95% intervals; filled markers are Holm significant. **(b)** Kendall's W for activation and optimiser in η_1, η_2 and the normalised deficit. **(c)** Architecture concordance within the MLP and CNN families. Conditioning amplifies the activation ordering, attenuates the optimiser ordering and changes the visibility of architecture differently in the two carriers.

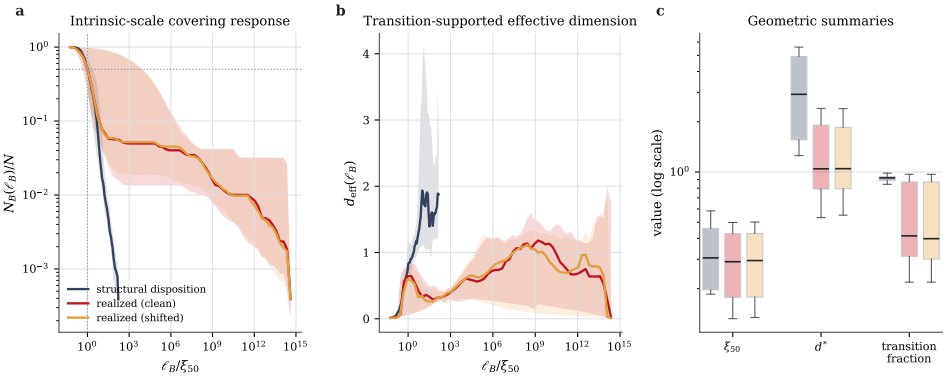


Fig. R18: Fashion-MNIST: intrinsic-scale covering response. **(a)** Median $N_B(\ell_B)/N$ against ℓ_B/ξ_{50} with interquartile bands. **(b)** Transition-supported d^F_{eff}. **(c)** Per-geometry ξ_{50}, d^* and transition fraction. The main structural-to-clean change is a collapse of transition support and crossover strength, with a small displacement of ξ_{50}. Every Fashion-MNIST geometry satisfies the credibility gate.

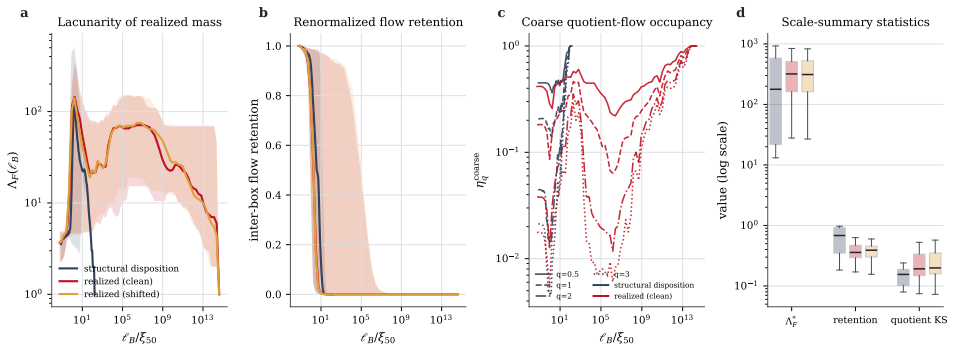


Fig. R19: Fashion-MNIST: multiscale texture and coarse flow. **(a)** Flow-mass lacunarity. **(b)** Inter-box flow retention. **(c)** Normalised quotient-flow Rényi occupancies for $q \in \{0.5, 1, 2, 3\}$. **(d)** Per-geometry peak lacunarity, retention and quotient-flow KS. Clean realisation is more lacunar and less inter-coupled than the structural disposition, and the Rényi panel shows a redistribution of coarse flow that does not follow from the box count alone.

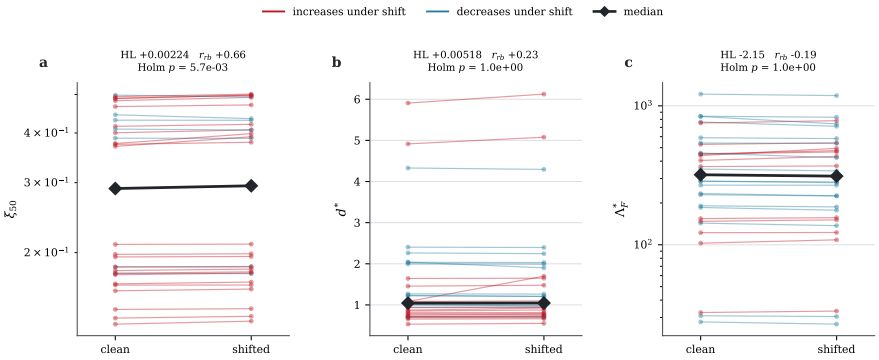


Fig. R20: Fashion-MNIST: paired geometric deformation under the 15 dB sensor shift. One line per geometry-cohort network joins its clean and shifted values; the heavy line is the median. Titles report the Hodges-Lehmann shift, rank-biserial effect and Holm-adjusted paired p. **(a)** ξ_{50} increases slightly but consistently. **(b)** d^* shows no corrected population shift. **(c)** Peak lacunarity is comparatively stable. A strong intrinsic response to the perturbation therefore need not deform every geometric summary.

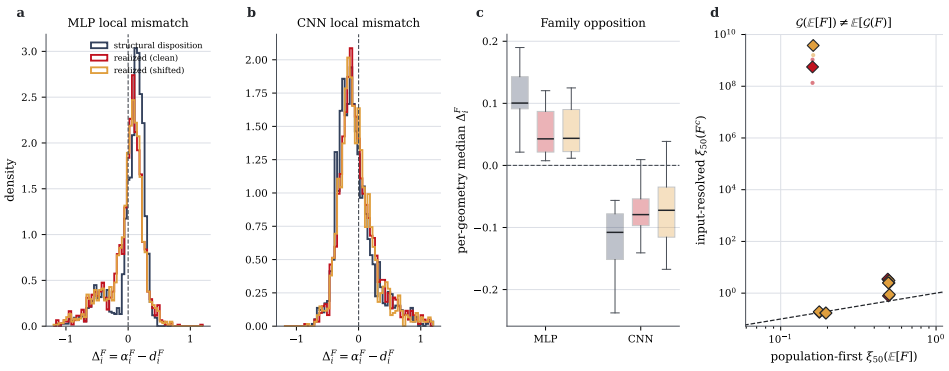


Fig. R21: Fashion-MNIST: carrier-resolved local geometry and non-commutation. **(a)** MLP node-level local support-mass mismatch. **(b)** CNN node-level mismatch. **(c)** Per-geometry local medians: positive for MLPs and negative for CNNs, both moving towards zero under clean realisation. **(d)** Input-resolved against population-first ξ_{50}, with within-cell input medians (diamonds) and the identity line. The extreme ReLU MLP displacement is read as sparse-flow metric stretching under the declared inverse-flow ruler, not as an absolute scale.

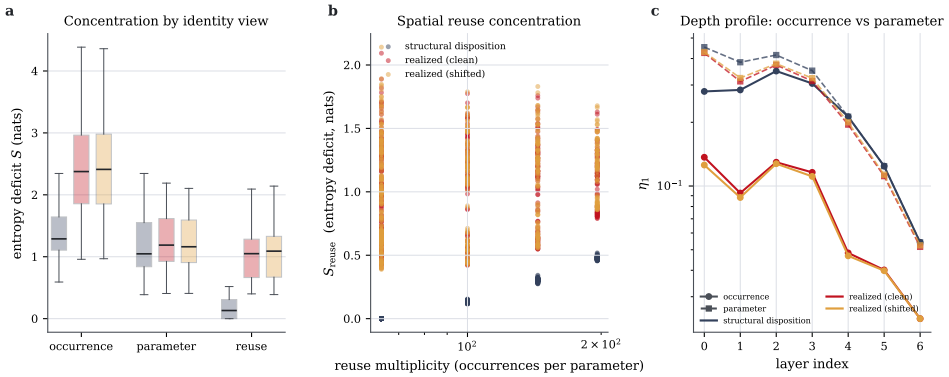


Fig. R22: Fashion-MNIST: occurrence, parameter-identity and spatial-reuse concentration in CNNs. **(a)** Entropy deficits for execution occurrence, parameter identity and residual reuse. **(b)** Reuse multiplicity against S_{reuse}. **(c)** Layerwise effective occurrence and parameter fractions. Clean conditioning concentrates spatial occurrence much more strongly than parameter identity, and the shift adds a small further reuse concentration. The S quantities are entropy deficits.

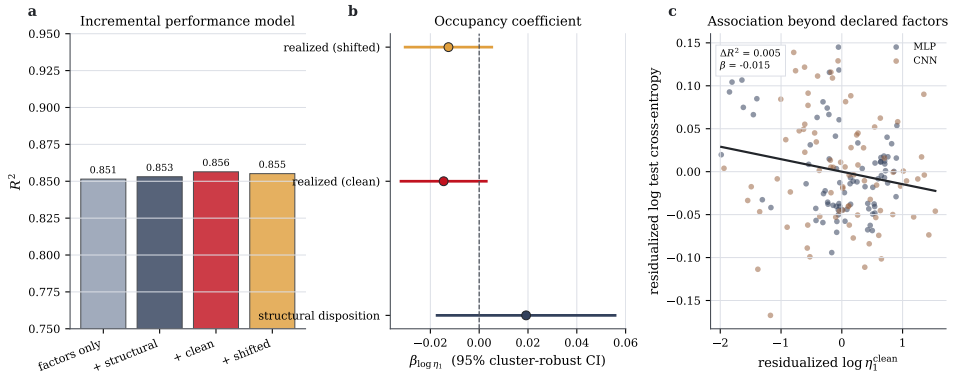


Fig. R23: Fashion-MNIST: post-hoc association between occupancy and classification loss. **(a)** Design-factor R^2 and the fits obtained by adding structural, clean or shifted $\log \eta_1$. **(b)** Occupancy coefficients with 95% cluster-robust intervals. **(c)** Partial association between residualised clean $\log \eta_1$ and residualised log test cross-entropy. Realised occupancy contributes only a small and statistically uncertain increment beyond the design factors.

6 Realisation-guided intervention

Question. Sections 2 to 5 establish that a condition realises a small and selective part of the disposition. A description of where realised flow goes does not yet show that the computation there matters for the response. We therefore intervene on the 324 frozen benchmark networks (Appendix E.13). An intervention multiplies the contribution of each selected carrier by $g = 1 - \alpha$ in both executions of the condition and recomputes every downstream state, with parameters and architecture unchanged. At equal budget and dose, a carrier set is more responsible for the response under c when its attenuation displaces that response more.

Design. Three layerwise sets of equal budget are compared. The realised set R contains the top tenth of positive realised support. The structural set S is selected from the structural measure towards the same output. The matched set M is drawn from the remaining positive realised support and matched on source-activation magnitude, so it controls for generic active computation. Forty balanced conditions per network are evaluated in the clean and shifted environments with $\alpha = 0.25$, giving 77,760 interventions. The operator suite passes 34 of 34 tests, and no intervention leaves the measure undefined (Table A.E42).

Realised flow locates load-bearing computation. Figure R24(i) shows that attenuating R displaces the response more than attenuating M in every network of all four dataset-environment populations (rank-biserial 1, Holm $p \approx 7.4 \times 10^{-28}$). The median advantage is 2.730 and 2.548 on clean and shifted Fashion-MNIST and 0.394 and 0.267 on clean and shifted MPI3D (Table A.E43a). The result does not rest on loose matching. With the mean matching distance restricted to 0.10, 88.3% of the conditions remain and 99.8% of the model-environment units keep R above M.

The activation sets the relative weight of disposition and realisation. R also exceeds S in most networks (Table A.E43a), and Figure R24(ii) shows that this comparison is organised by activation. The fraction of networks with R above S averages 0.940 under ReLU and 0.968 under SiLU but 0.148 under tanh. Under tanh, the median R minus S is negative in every dataset, environment and family, from -0.014 to -0.388 (Table A.E43b). Across the 24 activation-resolved strata, greater overlap of R and S accompanies a smaller realised advantage ($\rho_s \approx -0.61$, $p \approx 1.5 \times 10^{-3}$).

Re-realisation is ordered more uniformly than the output. Panel (iii) follows the measure after attenuation. Attenuating R or S broadens the surviving measure, while attenuating M sharpens it. The realised increase of η_1 exceeds the structural increase by factors from 1.46 to 1.76. The Hellinger displacement under R is about 0.54 on Fashion-MNIST and 0.64 to 0.65 on MPI3D, against 0.42 to 0.50 under S and 0.13 to 0.20 under M. Figure R25(i) relates this internal movement to the output at two scales. Across networks, larger internal displacement accompanies larger output displacement (ρ_s about 0.55 to 0.70 for R with families pooled). Within a network, the conditionwise coupling depends on the cell. It is positive for Fashion-MNIST CNNs (0.207 and 0.108), negative for Fashion-MNIST MLPs (-0.150 and -0.148) and near zero on MPI3D (Table A.E43c). R exceeds S in Hellinger displacement in 88.9% to 99.4% of networks, and in output displacement in 64.8% to 73.5%. There is thus no single conversion from movement of the measure to movement of one output coordinate.

Responsibility and task value are different coordinates. Figure R25(ii) follows the consequence further outward. The scale-free manifestation forms three separated bands, near 0.55, 0.33 and 0.08 for R, S and M. On Fashion-MNIST, R flips 13% of clean MLP and 33% of clean CNN predictions, against 2% and 6% for M. In shifted MPI3D CNNs, however, the median task-loss change is -0.0588 under R, -0.0279 under S and -0.0121 under M. Attenuating the carriers most responsible for the current response thus moves that response most and yet lowers the task loss. Panel (iii) adds a post-hoc ratio of relative manifestation to D_H, ordered R above S above M in all eight cells. A carrier can therefore participate strongly in realisation while the response it supports is poorly aligned with the external objective.

Transition. If realisation is conditional, the location of responsibility may move with the condition. Section 7 tests this directly.

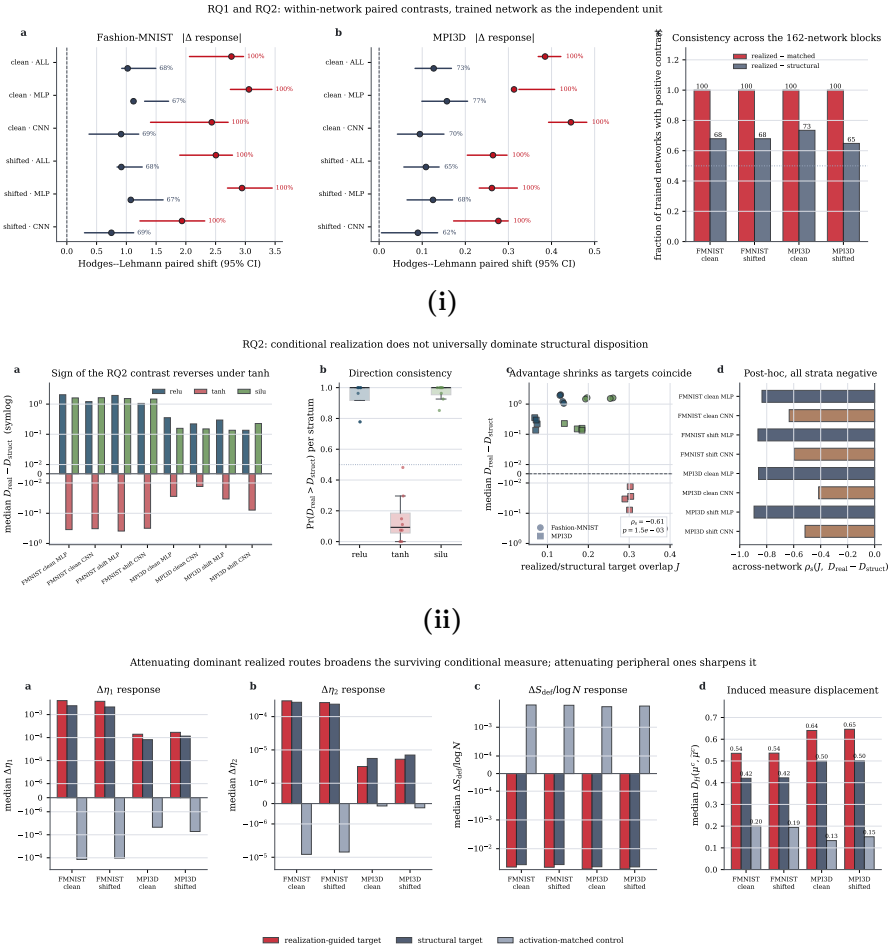


Fig. R24: Realisation-guided intervention: contrasts, activation and re-realisation. **(i)** (a,b) Hodges-Lehmann within-network shifts with 95% bootstrap intervals for realised minus matched (red) and realised minus structural (slate), labelled with the fraction of networks positive; (c) direction consistency of the two contrasts in each 162-network dataset-environment block. **(ii)** (a) Median realised minus structural displacement in each dataset-environment-family cell by activation: ReLU and SiLU favour the realised set and tanh reverses the ordering; (b) fraction of networks with $D_{\text{real}} > D_{\text{struct}}$ across the eight family-resolved strata; (c) post-hoc association between target overlap and the realised advantage; (d) across-network Spearman association between overlap and the realised minus structural displacement in every stratum. **(iii)** (a,b) Median changes of effective Shannon and order-two occupancy; (c) median change of normalised entropy deficit, where high-flow realised and structural attenuation broaden the surviving measure and matched peripheral attenuation sharpens it; (d) Hellinger displacement of the full normalised measure.

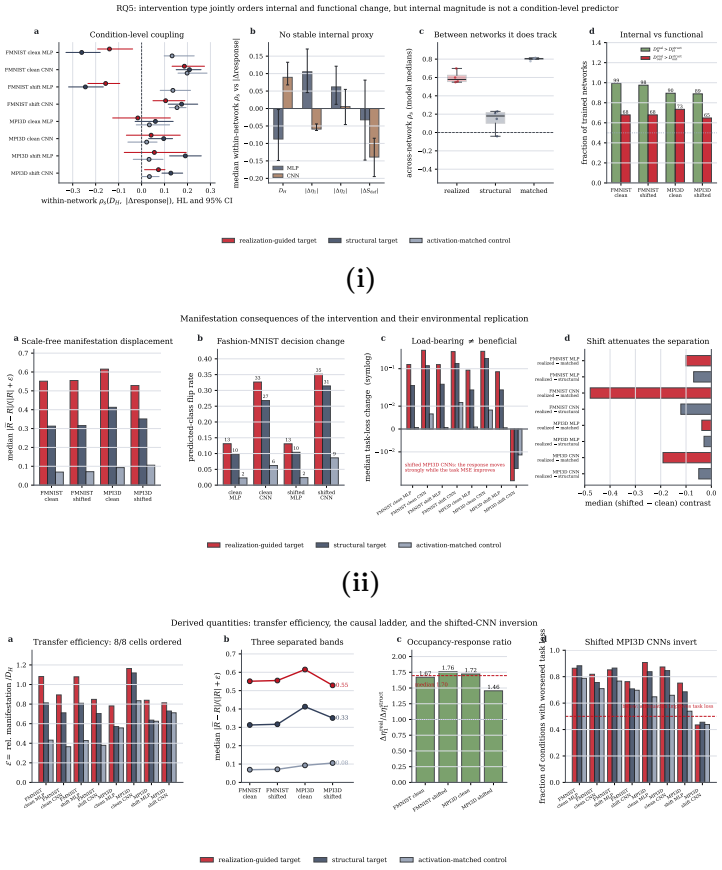


Fig. R25: Realisation-guided intervention: internal and external consequences. **(i)** (a) Within-network conditionwise Spearman coupling between Hellinger and functional displacement, with Hodges-Lehmann locations and bootstrap intervals for each set; (b) family-resolved summaries of several internal proxies under realised targeting, where family-task cells occupy different coupling regimes; (c) across networks, model-median internal and functional susceptibility are positively associated for all three sets; (d) realised targeting exceeds structural targeting far more universally in Hellinger displacement than in final functional displacement. **(ii)** (a) Dimensionless manifestation displacement of the three sets; (b) Fashion-MNIST prediction-flip rates; (c) median task-loss change by dataset, environment and family on a symmetric logarithmic scale, which retains the negative shifted MPI3D CNN effects; (d) shift-minus-clean change of the two causal separations. **(iii)** (a) Post-hoc transfer-efficiency ratio of relative manifestation to D_H, ordered realised above structural above matched in all eight cells; (b) separated bands of scale-free manifestation displacement; (c) ratio of realised to structural $\Delta\eta_1$, with median about 1.70; (d) fraction of conditions with worsened task loss, below 0.5 for shifted MPI3D CNNs.

7 Condition crossover

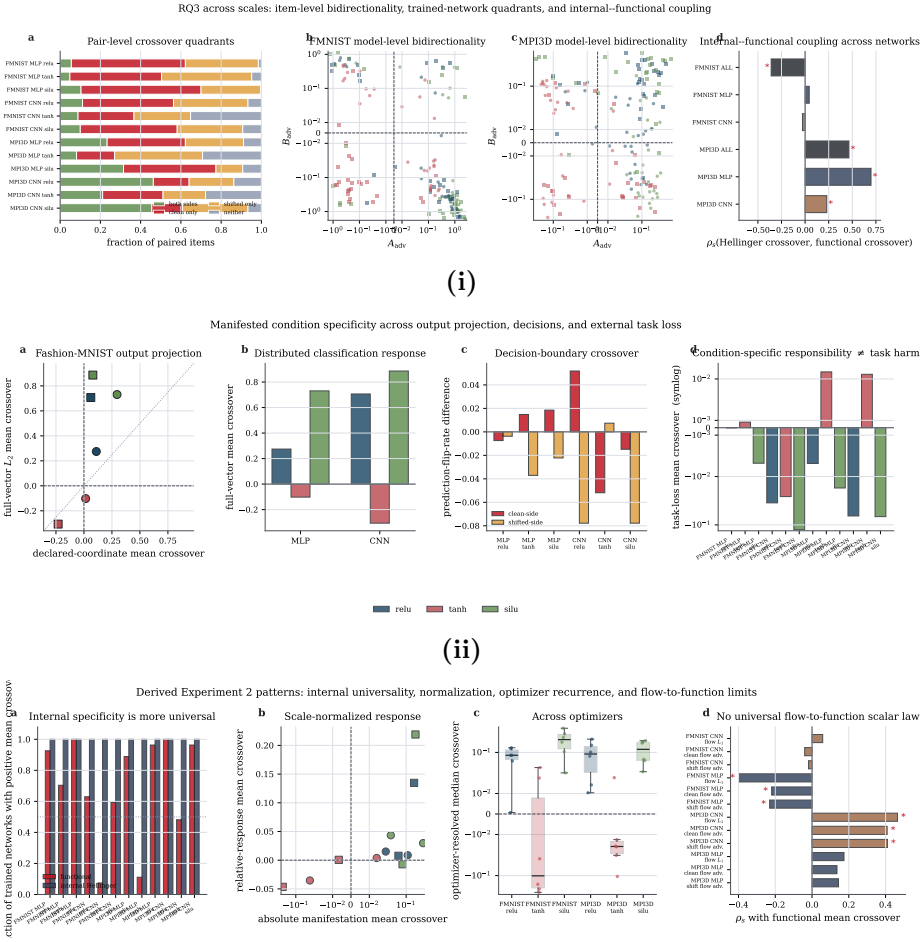
Question and design. We ask whether a set that is responsible under one condition is less responsible under another. Each network is evaluated on ten paired items, each a clean input and its 15 dB shifted version. For each carrier, the difference of its flow under the two conditions ranks it by the condition that realises it preferentially. Set $\mathcal{T}_A$ holds the carriers preferred under the clean condition, and set $\mathcal{T}_B$ those preferred under the shifted condition. Each set is attenuated under each condition with $\alpha = 0.25$, giving 12,960 interventions. The clean-side advantage compares the two sets under clean evaluation, the shifted-side advantage compares them under shifted evaluation, and their mean is the mean crossover. Each side therefore compares the origin of the target within one fixed evaluation condition. The conditions are well separated at baseline. The median own-to-other flow-mass ratio of the preferential sets is about 1.81 on Fashion-MNIST and 3.3 on MPI3D.

Each set reorganises its own condition more. At the level of the measure, the crossover holds in every network. The mean Hellinger crossover is positive in all networks of both benchmarks, with medians of 0.072 and 0.128 (Table A.E44c).

The output crossover depends on the benchmark and the activation. On Fashion-MNIST, the clean-side advantage is supported after correction (Holm $p = 6.70 \times 10^{-6}$) and the mean crossover is small and positive (Holm $p = 1.98 \times 10^{-3}$). On MPI3D, both sides are positive (Holm $p = 3.50 \times 10^{-6}$ and 1.23×10^{-3}), and so is their mean (Holm $p = 1.86 \times 10^{-10}$). ReLU and SiLU carry these effects. The mean crossover is significant for both in MLPs on both benchmarks and in MPI3D CNNs, where it reaches 0.166 and 0.181, while no tanh cell is significant (Table A.E44b).

The figure resolves the median into its parts. Figure R26(i) returns to single items. The median fraction of pairs with both sides positive is about 9.4% on Fashion-MNIST and 27.4% on MPI3D, and it reaches 46.3% for MPI3D CNN ReLU and 45.6% for CNN SiLU. The Fashion-MNIST asymmetry is thus assembled from one-sided item responses. Across MPI3D networks, the Hellinger and manifestation crossovers are positively associated ($\rho_s \approx 0.46$), whereas the pooled Fashion-MNIST association is negative and the family-separated ones are near zero. Panel (ii) changes the observed output. On Fashion-MNIST, the full output vector amplifies the ReLU and SiLU crossover. For CNN ReLU it rises from 0.061 on the declared coordinate to 0.707, and for CNN SiLU from 0.079 to 0.886 (Table A.E44d). Tanh stays negative in CNNs in both views. Prediction flips show a smaller and more heterogeneous crossover, as expected when a large logit movement can remain within one decision region. In MPI3D CNNs, the ReLU and SiLU manifestation crossovers of 0.166 and 0.181 accompany task-loss crossovers of about -0.065 and -0.067. Panel (iii) relates the crossover to the baseline separation of the conditions. In MPI3D CNNs, the mean crossover correlates with the baseline L_1 separation at $\rho_s \approx 0.47$, and in Fashion-MNIST MLPs at $\rho_s \approx -0.40$.

Interpretation. Condition specificity is most stable at the level of re-realisation. Its expression at the output depends on the benchmark, on the activation and on the projection of the output that is observed. The declared coordinate and the full vector are two projections of one re-realised computation, and the activation split persists in both.



(i)

(ii)

(iii)

Fig. R26: Condition crossover: item scale, output representation and derived patterns. **(i)** (a) Pair-level quadrants: both sides positive, clean side only, shifted side only or neither; (b,c) model-level clean-side against shifted-side advantage for Fashion-MNIST and MPI3D, with colour for activation and marker for family; (d) across-network Spearman coupling between Hellinger and manifestation mean crossover by dataset and family. **(ii)** (a) Fashion-MNIST mean crossover on the declared coordinate against the L_2 crossover of the full response vector; (b) full-vector crossover by family and activation; (c) difference in predicted-class flip rate between the condition-preferential and competing sets within a fixed evaluation condition; (d) mean crossover of task-loss change across family-activation cells. **(iii)** (a) Fraction of networks with positive functional against positive Hellinger mean crossover in every family-activation cell; (b) absolute against scale-normalised mean crossover; (c) optimiser-resolved crossover medians by dataset and activation; (d) across-network Spearman associations between functional mean crossover and three summaries of baseline flow differentiation.

8 Dose response

Question and design. A causal reading of responsibility predicts that a stronger attenuation of the same carriers displaces the response further. Each realised, structural and matched set is therefore selected once from the baseline state and held fixed. It is attenuated with $\alpha \in \{0.10, 0.25, 0.50\}$, so that it retains 90%, 75% and 50% of its contribution, under ten conditions in each environment. This gives 58,320 interventions, with the curves and their tests in Table A.E45.

Responsibility is graded. Both outcomes rise with dose in every population. For the realised set on clean Fashion-MNIST, the median manifestation displacement is 1.847, 3.834 and 5.895, and the Hellinger displacement 0.229, 0.531 and 0.843. On clean MPI3D, the manifestation displacement is 0.204, 0.480 and 0.783. Between 90.1% and 98.8% of networks have strictly increasing realised curves, and every Hellinger increment is positive in all networks. The dose intervals differ, so we compare curve shape through the local slopes s_{low} and s_{high}. The realised curve decelerates in a median of 96.3% of ReLU and 88.9% of SiLU networks, whereas 59.3% of tanh networks accelerate (Table A.E45c).

The two contrasts evolve differently. Figure R27(ii) shows that the realised minus matched advantage widens with dose, from 1.555 to 3.159 and 4.383 on clean Fashion-MNIST, and at least 96.9% of networks are positive at every dose in every population. The realised minus structural advantage behaves differently (panel iii). On clean Fashion-MNIST it is 0.950, 1.468 and 1.012, because the structural curve catches up at strong attenuation, with increments of 1.136 and 1.941. It also peaks at the middle dose on shifted Fashion-MNIST and on shifted MPI3D, whereas on clean MPI3D it still grows slightly (0.062, 0.116 and 0.136). Activation organises this contrast. On clean Fashion-MNIST, it is 1.166, 2.287 and 2.565 under ReLU, 1.364, 1.885 and 1.353 under SiLU, and -0.106, -0.439 and -1.105 under tanh.

Occupancy and displacement are different coordinates. Panel (i) shows that realised and structural attenuation broaden occupancy at every dose, while matched attenuation sharpens it. On clean Fashion-MNIST, the realised median $\Delta\eta_1$ moves from about 0.00081 to 0.00398 and back to 0.00319, while its Hellinger displacement keeps rising. Panel (iv) shows Fashion-MNIST prediction flips under clean realised attenuation rising from 7.5% to 21.8% and 46.8%, against 6.4%, 17.3% and 37.3% for the structural set and 2.0%, 4.9% and 11.4% for the matched set. Panel (v) follows the shifted MPI3D CNNs, in which realised attenuation lowers the task loss under ReLU at all three doses (-0.089, -0.187 and -0.102) and raises it under tanh at the highest dose ($+0.050$). Panel (vi) places the middle dose against the highest. At $\alpha = 0.25$, the realised set already reaches about 61% to 68% of the high-dose manifestation and 63% to 70% of the high-dose Hellinger displacement.

Interpretation. Responsibility is therefore graded and largely monotone, with diminishing returns under ReLU and SiLU. The advantage of the realised set over generic active computation grows with dose. Its advantage over the structural set depends on the dose, and the single-dose comparison of Section 6 lies at the largest value of that contrast on three of the four populations.

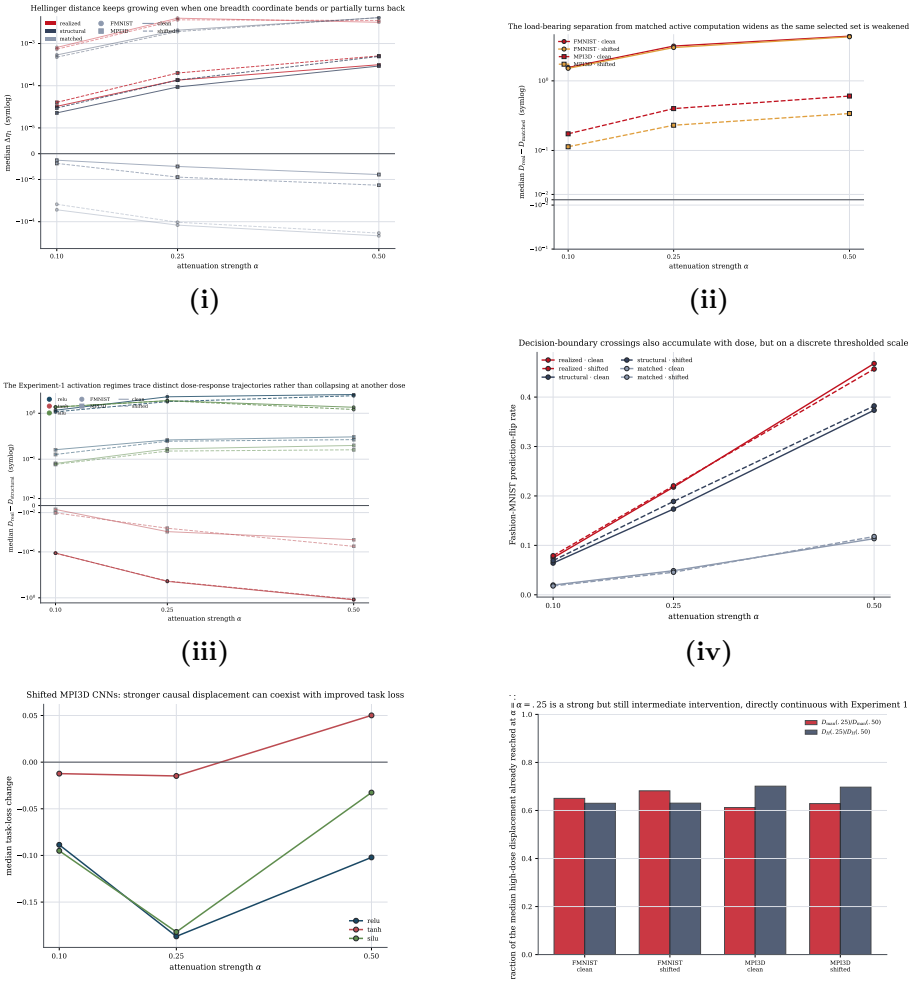


Fig. R27: Dose response: occupancy, contrasts, decisions and task loss. **(i)** $\Delta\eta_1$ across dose for each dataset, environment and set; high-flow realised and structural attenuation broaden occupancy and matched attenuation sharpens it, and on Fashion-MNIST the realised $\Delta\eta_1$ can peak near the middle dose while the Hellinger displacement keeps rising. **(ii)** Realised minus matched advantage, which grows with dose. **(iii)** Realised minus structural trajectories by activation: ReLU and SiLU stay on the realised side of zero and tanh on the structural side, often moving further from zero with dose; marker for dataset and line style for environment. **(iv)** Fashion-MNIST prediction-flip rates, which rise with dose for all three sets, highest for the realised set. **(v)** Shifted MPI3D CNN task-loss change under realised attenuation: ReLU and SiLU improve at all three doses, most near the middle dose, while tanh improves slightly and then degrades at the high dose. **(vi)** Fraction of the high-dose manifestation and Hellinger displacement already reached at $\alpha = 0.25$ for the realised set.

9 Multiscale geometry after intervention

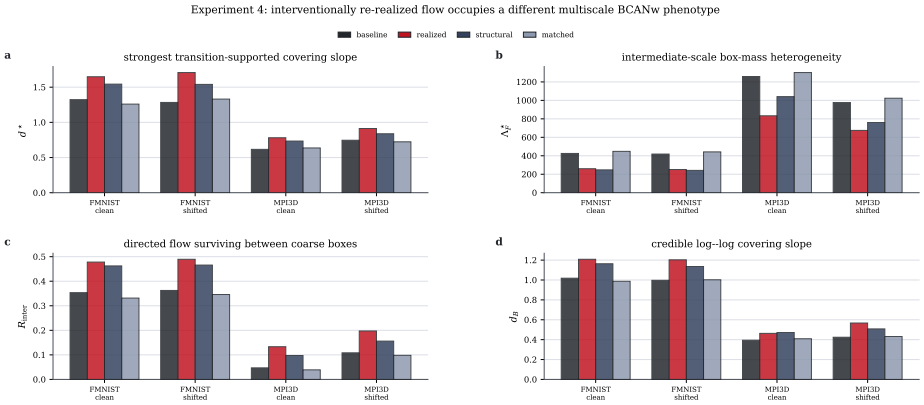
Question and design. The preceding sections follow the response and the measure after intervention. We now ask whether re-realisation also leaves a trace across scale. For each network and environment, the flows of ten balanced conditions are averaged carrier by carrier. The BCANw instrument is then applied to four states: the baseline and the states after realised, structural and matched attenuation at $\alpha = 0.25$. The instrument is identical across the four states, with 64 scales, the enrichment ruler with $p = 1$, undirected shortest paths and directed quotient flow. By the equivariance of BCANw under a common ruler, the differences between the states therefore arise from the realised flow. Ten descriptors are analysed, and Table A.E46 holds the state medians and the tests.

Re-realisation has a recurrent geometric backbone. Figure R28 shows that seven of the ten descriptors shift after realised attenuation in every dataset-environment population. Four of them form a common backbone across the 24 family-activation strata. The crossover height d^* rises in 16 strata, peak lacunarity falls in 21, inter-box retention rises in 20 and the credible d_B^F rises in 14 of 23, with no stratum moving against these directions. On clean Fashion-MNIST, for example, d^* rises from 1.323 to 1.648, peak lacunarity falls from 425.3 to 259.8 and retention rises from 0.353 to 0.478. The re-realised flow thus covers with steeper transitions, fills boxes more evenly and keeps more directed flow between them. The characteristic scale carries task-specific information. $\log \xi_{50}$ rises by 0.148 and 0.165 on Fashion-MNIST, falls by 0.201 on clean MPI3D and does not change significantly on shifted MPI3D.

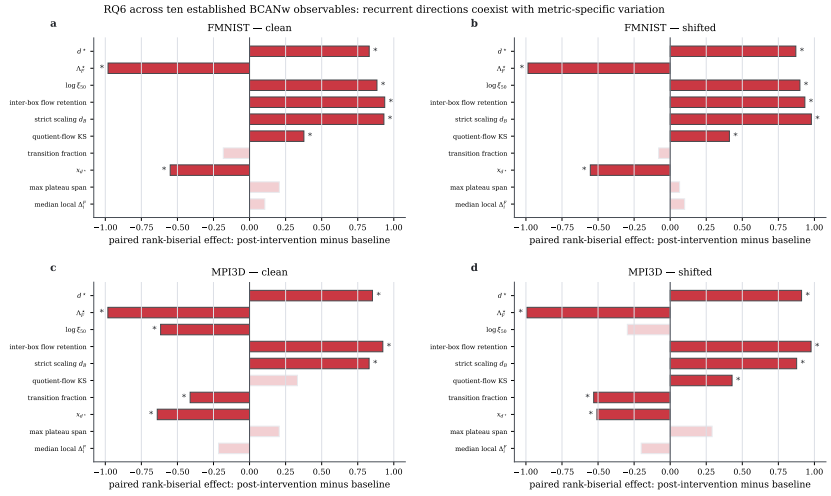
The geometric change is carrier specific. Figure R29(i) compares the absolute change under R with the changes under the controls. The realised change exceeds the matched change on 9 or 10 of the 10 descriptors in three populations and on 8 in shifted MPI3D, but it exceeds the structural change on only 4 or 5 (Table A.E46c). The geometric phenotype therefore separates realised from generic active computation more sharply than it separates realised from structural targeting. Panel (ii) resolves the activation. Tanh networks reorganise strongly. Lacunarity falls and retention rises significantly in all eight tanh cells, d_B^F rises in six and d^* in four. The activation thus decides which coordinates carry the change most consistently, while every activation changes the geometry.

Scale, coupling and validity. Figure R30(i) shows that the environment organises the MPI3D response. On MPI3D, the shift makes the realised increases of d^*, retention and d_B^F larger and the responses of $\log \xi_{50}$ and x_{d^*} less negative, whereas on Fashion-MNIST no environment contrast survives correction. Panel (ii) relates the geometric change to the Hellinger displacement across networks. The coupling is positive for the maximum plateau span in every population (0.44 to 0.63) and for peak lacunarity (0.26 to 0.43). For $\log \xi_{50}$ its sign follows the task, at -0.41 on clean Fashion-MNIST and $+0.42$ on clean MPI3D. Panel (iii) secures the comparison. With the matching distance restricted to at most 0.10, the realised change still exceeds the matched change in about 81% to 100% of networks for the five recurrent descriptors.

Interpretation. A carrier-level intervention leaves a coherent multiscale trace. The tanh result is especially informative. Its output ordering of R and S is reversed (Section 6), yet its geometry reorganises in the common direction, so functional



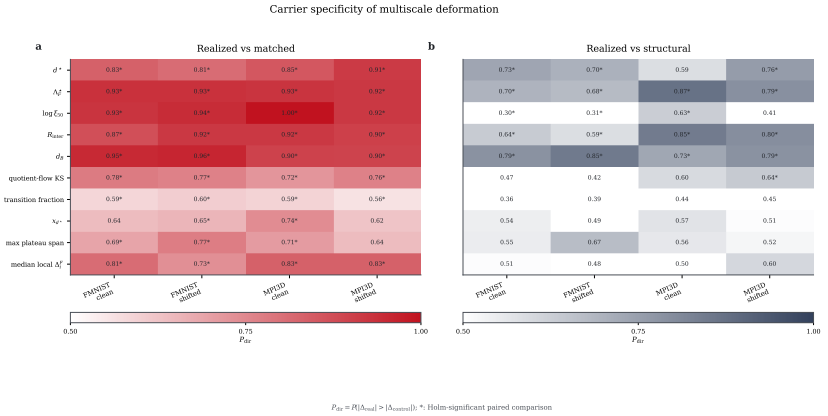
(i)



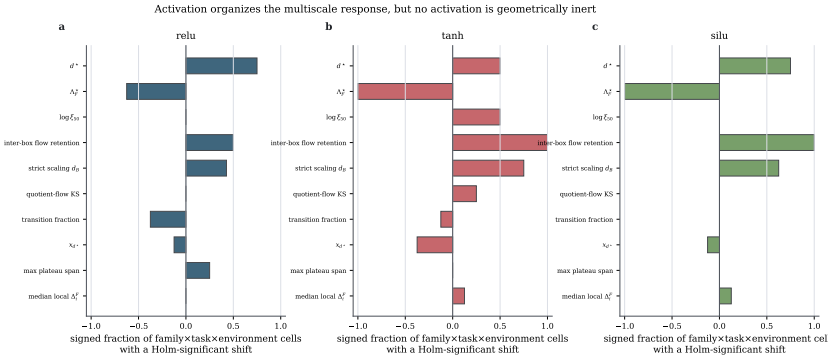
(ii)

Fig. R28: Multiscale geometry after intervention: states and primary effects. **(i)** Population medians of (a) d^*, (b) Λ_F^*, (c) mean directed inter-box retention and (d) credible d_B^F, for baseline (black) and after realised (red), structural (slate) and matched (grey) attenuation; the four horizontal settings are independent dataset-environment populations, not a time axis. **(ii)** Paired rank-biserial effects of realised attenuation against baseline for all ten descriptors in each population; saturated bars with an asterisk pass the global Holm correction, and each direction is read according to the meaning of its descriptor.

reversal and geometric reorganisation are distinct statistics of one intervention. The geometry is a representation of re-realisation under a declared ruler, and we do not promote it to the causal mechanism.

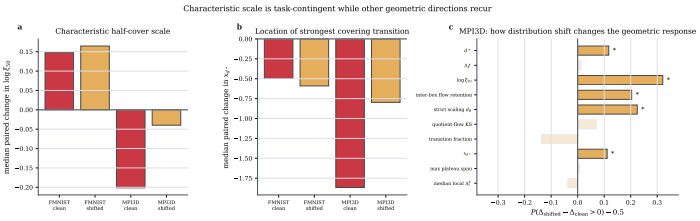


(i)

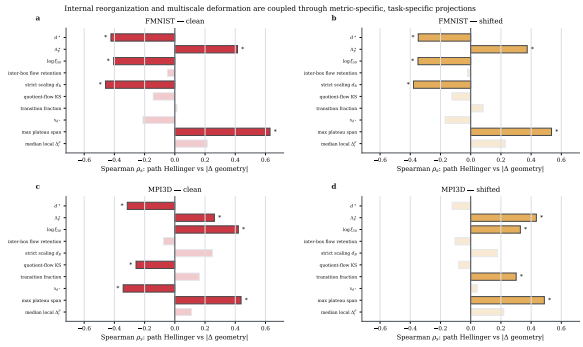


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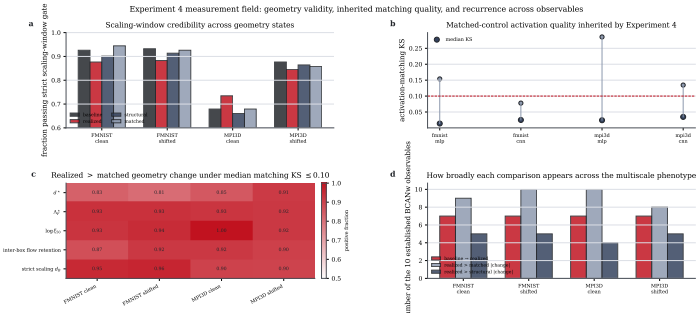
Fig. R29: Multiscale geometry after intervention: set specificity and activation. **(i)** Fraction of networks for which the absolute geometric change under realised attenuation exceeds (a) the matched change and (b) the structural change; asterisks mark Holm-significant paired comparisons, and darker cells mean greater directional consistency. **(ii)** For each activation and descriptor, the signed fraction of the eight family-task-environment cells with a Holm-significant realised shift; positive bars are recurrent increases and negative bars recurrent decreases.



(i)



(ii)



(iii)

Fig. R30: Multiscale geometry after intervention: scale, coupling and validity. **(i)** (a) Median paired change of $\log \xi_{50}$; (b) median paired change of the normalised location x_{d^*} of the strongest transition; (c) for MPI3D, directional consistency of the shift-minus-clean difference of each realised geometric response, with asterisks for Holm-significant environment contrasts. **(ii)** Across-network Spearman correlations between Hellinger displacement and the absolute change of each descriptor, clean in red and shifted in ochre, with asterisks for post-hoc Holm significance. **(iii)** (a) Fraction of each geometry state with a credible scaling window; (b) median and 90th-percentile activation-matching distances inherited from the matched control; (c) fraction of networks with a larger realised than matched change after restricting to matching distance at most 0.10, for five recurrent descriptors; (d) number of the ten descriptors with a significant baseline shift, realised-over-matched advantage and realised-over-structural advantage.

10 Occurrence targeting and equivariance in CNNs

Question. Sections 3 to 5 show that conditioning moves CNN specialisation from parameter identity to spatial reuse. We now ask the corresponding causal question. Knowing which shared weight carries realised flow may not be enough to know where the present computation is vulnerable to intervention.

Design. A convolutional coefficient is used at every compatible position, so one shared identity has many occurrences. Occurrence-resolved targeting ranks occurrences by their realised flow. Shared-parameter targeting ranks identities by their summed flow and attenuates every reuse of a selected identity. The budget is matched per layer in primitive occurrences, and each selected term is multiplied by 0.75. The two contrasts are tested over the 81 CNNs of each dataset-environment cell, with Holm correction over the eight primary tests (Table A.E47a). Field A is the input-origin marginal of the realised flow, and Field B projects the selected first-layer occurrence flow to image coordinates. Both fields are diagnostics and not the intervention.

The spatial use of a shared weight matters. On Fashion-MNIST, occurrence targeting moves the measure further than shared-parameter targeting, with a median ΔD_H of 0.037 in both environments ($r_{rb} = 0.720$ and 0.692, Holm $p \leq 4.4 \times 10^{-7}$). On MPI3D, the difference is not significant. The occurrence set captures more baseline realised flow in every model-level block, with all-layer advantages from 1.477 to 2.101 (Table A.E47b). In the first convolution it touches about 135 to 158 of the 256 input positions, against 250 to 253 for shared-parameter targeting. Figure R31(i) makes this visible. The occurrence set is typically narrower and yet captures more realised flow, because it spends the budget on the spatial uses that carry the present computation.

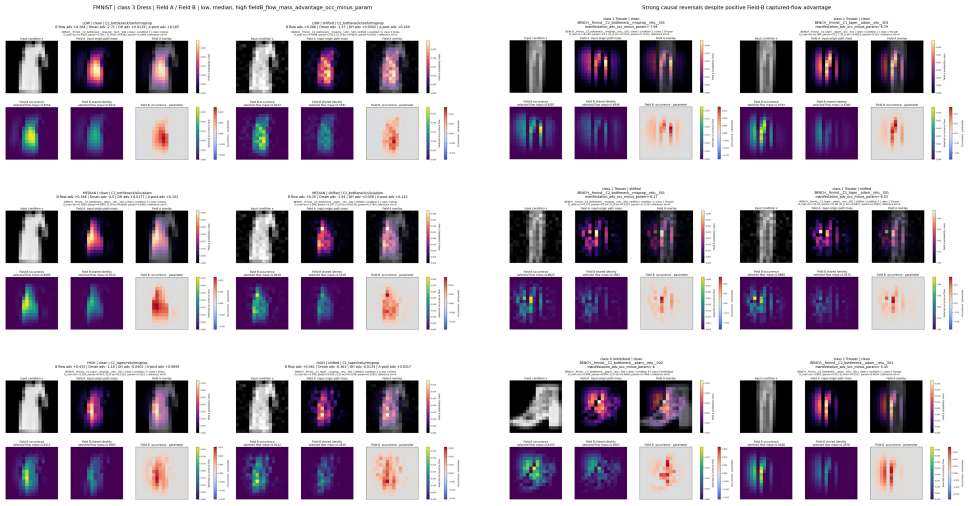
Locating the computation does not fix the size of the output change. The output effect follows activation. Under shifted evaluation on Fashion-MNIST, the manifestation advantage of occurrence targeting is +0.451 under ReLU, +0.319 under SiLU and −0.376 under tanh. Figure R31(ii) collects direct counterexamples, in which occurrence targeting captures more realised flow while shared-parameter targeting changes the final response more. Once the selected contributions are attenuated, the disturbance passes through the rest of the nonlinear network, where downstream routes can reinforce, cancel, saturate or compensate it. Quotienting occurrences to identities nevertheless discards information that matters for intervention.

Equivariance of the realised field. We then ask whether the realised input-origin field moves with the input. For a transformation T_g, the recomputed field $P_{T_g x}$ is compared with the transported field $(T_g)_{\#}P_x$ and with the untransformed field P_x, through the Hellinger errors H_{eq} and H_{inv}. The advantage $A_{\text{eq}} = H_{\text{inv}} - H_{\text{eq}}$ is positive when transport is the better description. Fashion-MNIST is evaluated under translations, rotations and scalings, and MPI3D under vertical translations, giving 32,400 comparisons. The identity transformation returns numerically zero error.

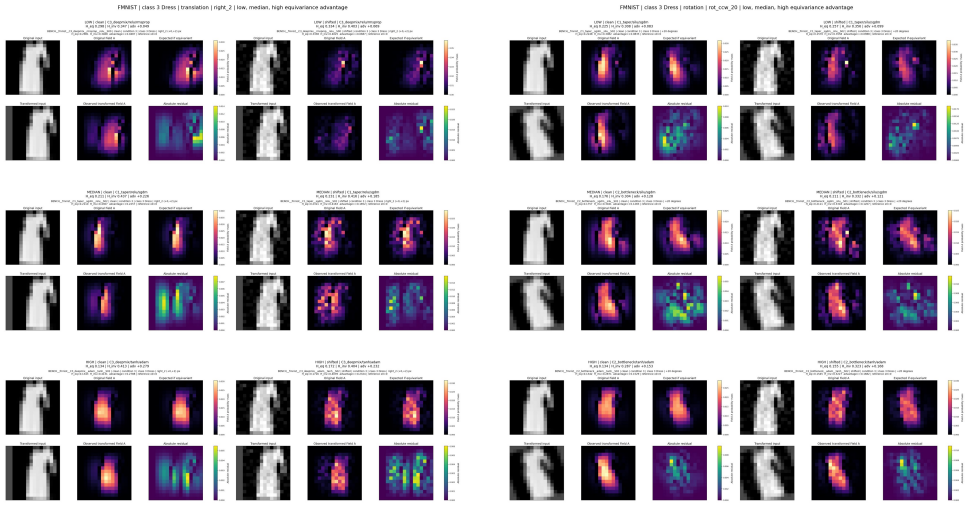
Translation and rotation are robust, and enlargement marks the boundary. Figure R31(iii) and (iv) show that the field follows translation and rotation. Transport is favoured in every network, with median advantages of 0.161 and 0.187 for a two-pixel right translation and 0.147 and 0.177 for an anticlockwise rotation of 20° on clean and shifted Fashion-MNIST (Table A.E47d). Figure R32 shows where this weakens. Shrinking to 0.90 still favours transport (0.038 and 0.055), whereas enlargement to 1.10

gives -0.006 on clean Fashion-MNIST, with 34.6% of networks positive, and $+0.009$ under the shift (panel i). On MPI3D, the field follows a two-pixel downward translation in both environments, with advantages of 0.497 and 0.364 (panel ii). The shift changes how strongly the field follows the transformation, and the direction depends on the dataset. On Fashion-MNIST, the advantage rises by about 0.058 for translation, 0.023 for rotation and 0.011 for scale, whereas on MPI3D it falls by 0.127. Panel (iii) shows the networks with the smallest advantage, in which the field still moves partly in the expected direction while transport no longer captures the realised redistribution by a large margin.

Interpretation. The realised computation of a trained CNN is approximately equivariant under translation and rotation, and the approximation weakens under enlargement. A positive advantage is not exact equivariance, which would require $H_{\mathrm{eq}} = 0$. The exact symmetry results of the paper concern carrier-preserving maps, and input transformations of a trained CNN are not of this kind. The worst-case audit therefore bounds the result to a population-level description.

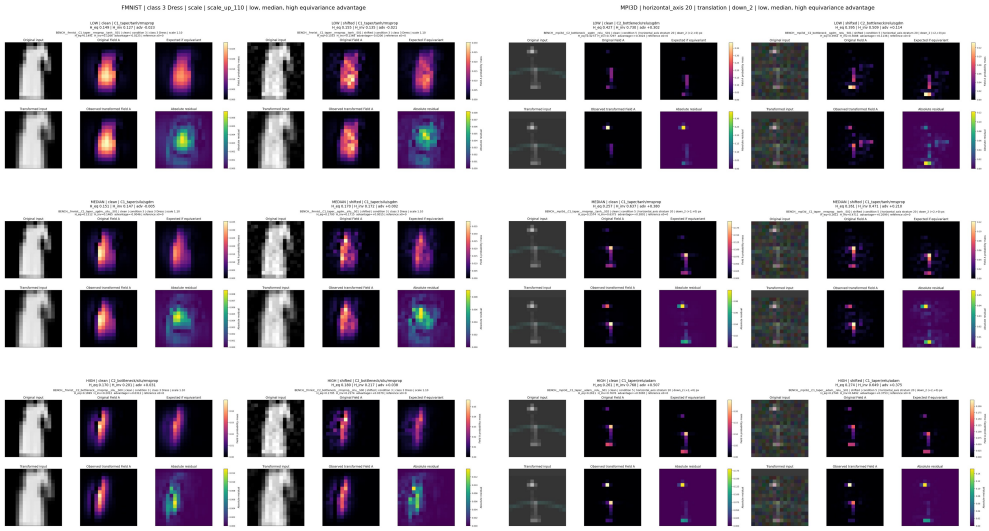


(i) **(ii)**

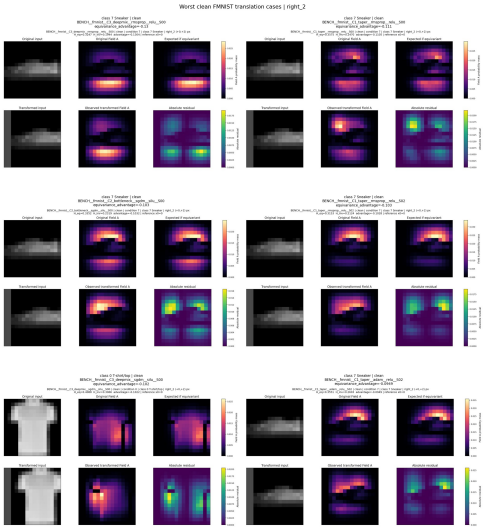


(iii) **(iv)**

Fig. R31: Spatial audits on Fashion-MNIST class 3 (dress) across networks. **(i)** Low, median and high cases ranked by Field-B flow advantage, in clean and shifted environments. Field A shows the input-origin realisation, and Field B the first-layer support of the occurrence-resolved and shared-parameter sets; the occurrence set is typically narrower yet captures more realised flow. **(ii)** Cases in which occurrence targeting captures more realised flow, whereas shared-parameter targeting changes the final response more. **(iii)** Two-pixel right translation, with cases ranked by equivariance advantage: observed field, transported prediction and residual. **(iv)** Rotation by 20° anticlockwise; the field generally rotates with the input, up to network-specific residuals.



(i) (ii)



(iii)

Fig. R32: Boundaries of equivariance. **(i)** Fashion-MNIST class 3 under enlargement by 1.10, the principal weak-transport regime: low-ranked cases can favour the unchanged baseline, while higher-ranked cases recover a positive advantage. **(ii)** MPI3D horizontal-position stratum 20 under a two-pixel downward translation: the field follows the translated scene in both environments, with a smaller advantage under the shift. **(iii)** The clean Fashion-MNIST networks with the smallest advantage for the two-pixel right translation: the field still moves partly in the expected direction, but transport no longer captures the realised redistribution by a large margin.